KI-Agenten in der Medizin

Scheitert der Fortschritt an der Zulassung?

Dr. med. Oscar Freyer – 5. November 2026

Interessenkonflikte

Honorare

ESMO

Springer Nature (Associate Editor, npj Digital Medicine)

Beratung

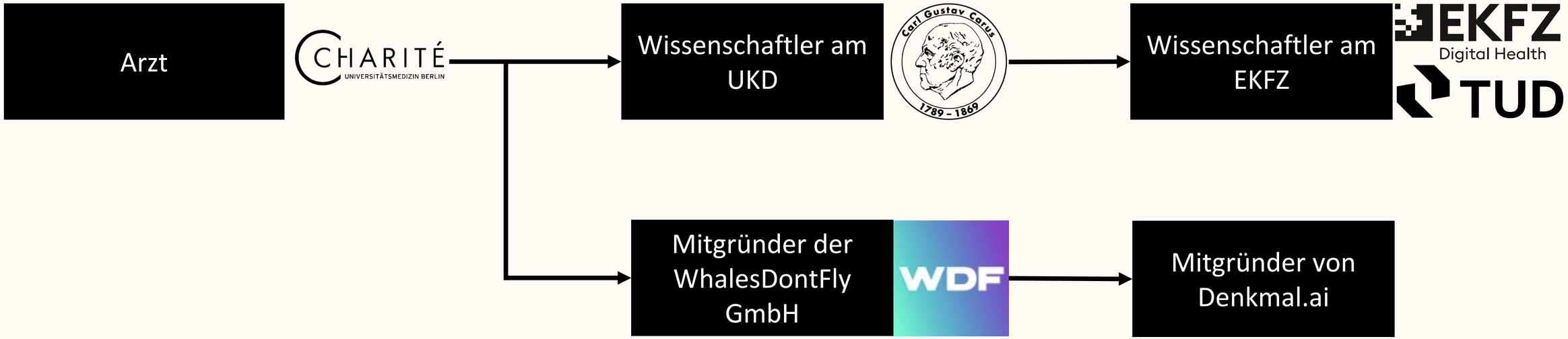
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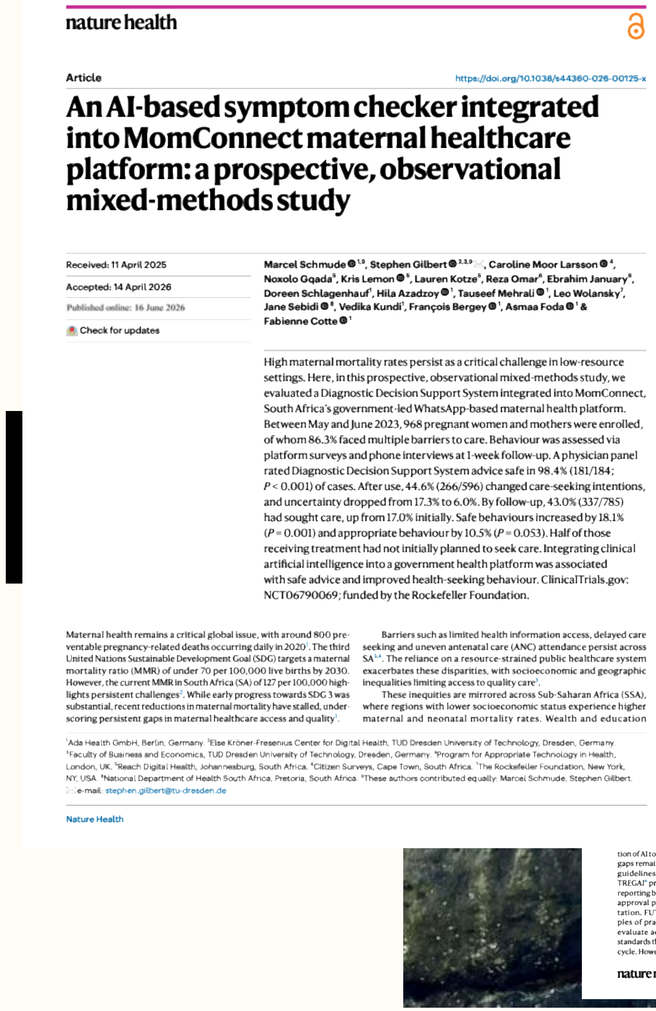
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Über mich


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Über mich



nature health

Article

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An AI-based symptom checker integrated into MomConnect maternal healthcare platform: a prospective, observational mixed-methods study

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Check for updates

Marcel Schmude[1,9], Stephen Gilbert[2,3,9] ✉, Caroline Moor Larsson[4], Noxolo Gqada[5], Kris Lemon[6,5], Lauren Kotze[5], Reza Omar[6], Ebrahim January[6], Doreen Schlagenhauf[7], Hila Azadzoy[1], Tauseef Mehrali[1], Leo Wolansky[7], Jane Sebidi[8], Vedika Kundi[1], François Bergey[1], Asmaa Foda[1] & Fabienne Cotte[1]

High maternal mortality rates persist as a critical challenge in low-resource settings. Here, in this prospective, observational mixed-methods study, we evaluated a Diagnostic Decision Support System integrated into MomConnect, South Africa's government-led WhatsApp-based maternal health platform. Between May and June 2023, 968 pregnant women and mothers were enrolled, of whom 86.3% faced multiple barriers to care. Behaviour was assessed via platform surveys and phone interviews at 1-week follow-up. A physician panel rated Diagnostic Decision Support System advice safe in 98.4% (181/184; $P < 0.001$) of cases. After use, 44.6% (266/596) changed care-seeking intentions, and uncertainty dropped from 17.3% to 6.0%. By follow-up, 43.0% (337/785) had sought care, up from 17.0% initially. Safe behaviours increased by 18.1% ($P = 0.001$) and appropriate behaviour by 10.5% ($P = 0.053$). Half of those receiving treatment had not initially planned to seek care. Integrating clinical artificial intelligence into a government health platform was associated with safe advice and improved health-seeking behaviour. ClinicalTrials.gov: NCT06790069; funded by the Rockefeller Foundation.

Maternal health remains a critical global issue, with around 800 preventable pregnancy-related deaths occurring daily in 2020[1]. The third United Nations Sustainable Development Goal (SDG) targets a maternal mortality ratio (MMR) of under 70 per 100,000 live births by 2030. However, the current MMR in South Africa (SA) of 127 per 100,000 highlights persistent challenges[2]. While early progress towards SDG 3 was substantial, recent reductions in maternal mortality have stalled, underscoring persistent gaps in maternal healthcare access and quality[3].

Barriers such as limited health information access, delayed care seeking and uneven antenatal care (ANC) attendance persist across SA[3,4]. The reliance on a resource-strained public healthcare system exacerbates these disparities, with socioeconomic and geographic inequalities limiting access to quality care[3].

These inequities are mirrored across Sub-Saharan Africa (SSA), where regions with lower socioeconomic status experience higher maternal and neonatal mortality rates. Wealth and education

[1]Ada Health GmbH, Berlin, Germany. [2]Else Kröner-Fresenius Center for Digital Health, TUD Dresden University of Technology, Dresden, Germany. [3]Faculty of Business and Economics, TUD Dresden University of Technology, Dresden, Germany. [4]Program for Appropriate Technology in Health, London, UK. [5]Reach Digital Health, Johannesburg, South Africa. [6]Citizen Surveys, Cape Town, South Africa. [7]The Rockefeller Foundation, New York, NY, USA. [8]National Department of Health South Africa, Pretoria, South Africa. [9]These authors contributed equally: Marcel Schmude, Stephen Gilbert. ✉e-mail: stephen.gilbert@tu-dresden.de

Nature Health

Viewpoint

A future role for health applications of large language models depends on regulators enforcing safety standards

Oscar Freyer, Isabella Catharina Wiest, Jakob Nikolas Kather, Stephen Gilbert

Among the rapid integration of artificial intelligence in clinical settings, large language models (LLMs), such as Generative Pre-trained Transformer-4, have emerged as multifaceted tools that have potential for health-care delivery, diagnosis, and patient care. However, deployment of LLMs raises substantial regulatory and safety concerns. Due to their high output variability, poor inherent explainability, and the risk of so-called AI hallucinations, LLM-based health-care applications that serve a medical purpose face regulatory challenges for approval as medical devices under US and EU laws, including the recently passed EU Artificial Intelligence Act. Despite unaddressed risks for patients, including misdiagnosis and unverified medical advice, such applications are available on the market. The regulatory ambiguity surrounding these tools creates an urgent need for frameworks that accommodate their unique capabilities and limitations. Alongside the development of these frameworks, existing regulations should be enforced. If regulators fear enforcing the regulations in a market dominated by supply or development by large technology companies, the consequences of layperson harm will force belated action, damaging the potentiality of LLM-based applications for layperson medical advice.

Lancet Digit Health 2024; 6: e663–72

Else Kröner Fresenius Center for Digital Health, TUD Dresden University of Technology, Dresden, Germany (O Freyer, I C Wiest Dr med, Prof J N Kather Dr med, Prof S Gilbert PhD); Department of Medicine, Medical Faculty Mannheim, Heidelberg University, Mannheim, Germany (I C Wiest); Department of Medicine, University Hospital Dresden, Dresden, Germany (Prof J N Kather); Medical Oncology, National Center for Tumor Diseases, University Hospital Heidelberg, Heidelberg, Germany (Prof J N Kather)

Correspondence to: Prof Stephen Gilbert, Else Kröner Fresenius Center for Digital Health, TUD Dresden University of Technology, Dresden 01307, Germany
stephen.gilbert@tu-dresden.

Introduction

The use of artificial intelligence (AI) in clinical settings has increased since 2015, with various applications being used for specific purposes, such as medical imaging, patient care, or emergency medicine.[1] Since 2022, multimodal analysis, including radiology images.[2,3] The performance of individual LLMs and LLM-based systems, such as GPT-3.5, GPT-4, ChatGPT, Articulate Medical Intelligence Explorer, and Llama 2, was also measured in application scenarios for layperson end users, showing er a simple internet ns when compared ever, validating and l and safe clinical are in an era where



ndence

https://doi.org/10.1038/s41591-025-03787-4

ational partnership for governing tive artificial intelligence models icine

Check for updates

ve artificial intelligence models, such as genera- ersarial networks (GANs) sformer-based large lan- odels (LLMs), are devel- ed pace and positioned o clinical workflows and across the world. How- of GenAI in medicine and not just unprecedented lso systemic risks in the w technology and critical ms of safety, governance ight. GenAI and LLMs are n nature, possess broad ilities, and display evolv- ese characteristics chal- regulatory frameworks listic, task-specific artifi- models, such as those for al Device (SaMD).

amental risks associated ls applications in health- et to be fully addressed ory framework ('known as other risks and cha- even surfaced ('unknown unknowns include a lack aining data (including the hetic data for training'), s, hallucination of incor- t, and potential misuse in settings' (Box 1).

challenges, a variety of es and frameworks have ovide guidance on the rting and implementa-

tion of AI tools in medicine. However, notable gaps remain (Fig. 1). For example, reporting guidelines such as the TRIPOD-LLM[7] and TRIGAI[8] promote transparency in research reporting but do not extend to policymaking, approval processes, and actual implementation. FUTURE-AI[9] and FURM[10] are examples of practical frameworks to guide and evaluate adherence to best practices and standards throughout the entire AI device lifecycle. However, most provide only high-level recommendations without granularity in the technical details needed for specific operations.

Existing regulations such as the European Union (EU) medical device rules were not designed to address issues unique to GenAI and LLMs. First, the qualification of LLM-based applications as medical devices remains unclear or ill-defined. Risk classification and current degrees of control are unable to cater for risks specific to LLMs. Second, traditional methods used to evaluate SaMD generalize poorly to LLM-based applications. Although studies have demonstrated strong LLM performance in structured tasks such as medical question-answering, the application of LLM in real-world clinical decision-making is unlikely to be limited to such tasks. Third, safety and



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egulatory barriers to the on of AI agents in healthcare

, Jakob N. Kather & Stephen Gilbert

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ptive oversight

022, there has been a broad language models (LLMs) igence (GenAI). This trend e development of several nd their implementation facts have a clear medical n, may qualify as medical h standards set by medical e sparked a debate among ability of such products nt approvals, such as that the EU, demonstrate that tonomy can be approved, are adequately mitigated. ion to healthcare regula- become fully autonomous . These AI agents or agen- ly executing and control- sers with a high degree of ultiple components, such ols (for example, applica- guidance and the retrieval), which could be orches- handle decision-making, pletion' [7]. Because several n of a medical purpose, AI dical devices and thus be ever, as the fundamental nous processes and deci- nitor and model, at least potential, the regulation tain, an be approved as medi- ameworks in the USA and ed to be adapted to facili-

the jurisdiction and the application's specific medical purpose. In the healthcare sector, there are two broad types of AI-enabled applications: applications with a non-medical purpose, such as wellness apps and administrative applications (for example, simple documentation software), and applications with a medical purpose, such as clinical decision support systems, therapeutic applications and diagnostic tools.

Applications with a medical purpose often qualify as medical devices and must comply with medical device-specific laws, such as the EU Medical Device Regulation or the US Federal Food, Drug & Cosmetic Act. Medical devices have to follow clearly defined, highly regulated pathways to market, such as 510(k) submission, de novo classification or full premarket approval in the USA. In addition to medical device–specific laws, AI-enabled medical devices are also subject to general legislation, for example, privacy laws (GDPR (EU), HIPAA (USA), and so on) and AI-specific laws (such as the EU AI Act).

Since the mid-2010s, an increasing number of AI-enabled medical devices have received approval in the EU and the USA, most of which have a narrowly defined intended purpose and limited capabilities. Based on historical decisions, two aspects of the intended purpose may be identified as important for approvability: the scope of the intended purpose (narrow versus broad) and the degree of autonomy of the system[4] (Fig. 1).

Narrow and non-autonomous AI-enabled health applications are approvable under current regulations and pathways[4], albeit with some difficulties. This also applies to some GenAI applications, as demonstrated by the first approval of narrow and non-autonomous GenAI applications in the EU and the UK. Even individual narrow near-autonomous and autonomous systems that are not GenAI have been approved, although they often fall into high-risk classes[4].

By contrast, the approval of broad AI systems is more problematic, as current regulatory frameworks are designed for products that have a narrow scope, are not adaptive, keep human oversight and do not evolve after their placement on the market[4,9]. Such frameworks lack applicability to meaningful GenAI applications, as those tend to have a broader scope due to the general-purpose capabilities of the underlying model. AI agents built with GenAI models are not only broad in scope but, by definition, also possess a high degree of autonomy in their operation, with the ability to respond and produce a wide range of inputs and outputs for which they have not been specifically designed (Fig. 1). These characteristics result in a range of benefits and risks that are different in their nature from those of previous applications[4] (Table 1).

Overall, it can be expected that AI agents with an intended use that qualifies them as medical devices face considerable hurdles to regulatory approval. Systems with a narrowly defined scope and low degree of autonomy have a more straightforward path to approval, as they resemble on-market, non-agentic AI tools. However, the deployment of 'true' AI agents, which are both autonomous and broad in scope, is likely to remain impossible without substantial changes to current regulatory frameworks.

Regulation under current regulatory frameworks

The regulatory status of AI applications is highly dependent on geography and the use case, meaning that different laws may apply based on

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Brief communication

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The landscape of artificial intelligence-enabled medical devices in the EU and the US intended for intensive care units

Check for updates

Oscar Freyer[1] ✉, Stephan Buch[1], Adel Bassily-Marcus[2], Sven Zenker[3,4,5], Brian W. Pickering[6], Max Ostermann[1], Anett Schönfelder[1] & Stephen Gilbert[1,7]

Artificial intelligence (AI) is increasingly used in healthcare, yet translation into routine intensive care units (ICU) practice remains slow. Through a multimethod search, we identified 36 on-market ICU-specific AI-enabled medical devices in the US and EU, challenging previous research findings. Most devices focus on prediction. However, as availability does not equal proven benefit, adoption will depend less on the availability of additional models and more on addressing persistent implementation barriers.

Clinical research on artificial intelligence (AI) in healthcare has grown rapidly, covering traditional AI technologies, such as machine learning, as well as the latest innovations, such as Large Language Models (LLMs)[1]. In intensive and critical care, research focuses on diagnostic capabilities[2], disease prediction[3], therapy[4], as well as ICU administration, e.g., for optimal resource allocation[5], discharge planning[6], and generation of discharge letters[7].

Although this growing body of research is driven by the goal of improving patient care, its clinical relevan routine ICU practice. This integration is oft to Berkhout et al.[1], AI models for ICU u development or initial clinical validation sta to clinical integration and none reaching fu Lee et al.[8] identified a low number of AI-en available on the US market specifically des researchers describe the use of such comm

This discrepancy between academic translation and availability of authorized de the current evidence base: it remains uncle (ICU-AIeMD) have moved from the resea same time, there is a gap between regulat clinical utility, leading to limited integrati slowing widespread adoption[1,8]. Additional primarily focus on the US, supported by umentation and curated device lists, where dation remains lacking[1]. This gap, however apply structurally different regulatory fram

classification schemes, conformity assessment routes, and transparency requirements[12]. A detailed overview of the regulatory context is provided in the Supplementary Material. These regulatory differences may shape both the types of devices that reach the market and the evidence publicly available about them. Therefore, this study aims to (1) systematically identify commercially available ICU-AIeMD in the US and the EU, and (2) characterize and compare their regulatory and functional features to support understanding of the translation from research into authorized products across

[1]Else Kröner Fresenius Center for Digital Health, New Haven Health System, New Haven, CT, U Bonn, Germany. [2]Department of Anesthesiol Informatics, and Epidemiology, University Ho NY, USA. [7]Faculty of Business and Economi

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NEJM AI

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ORIGINAL ARTICLE

From Advice to Action — Real-World Behavior of Patients Using an Integrated Diagnostic Decision Support System for Navigating the Health Care System

Fabienne Cotte, M.D.,[1] Filipa Dias Lourenço, M.D.,[2] Miguel Paiva Pereira, M.D.,[2] Nisha Kini, M.B.B.S., M.P.H.,[1] Marcel Schmude, M.D.S.,[1] Andreia Pimenta, Ph.D.,[1] Athena Lemesiou, Ph.D.,[1] Stephen Gilbert, Ph.D.,[3,4] Tauseef Mehrali, M.B.Ch.B. (Hons.),[1] Micaela Seemann Monteiro, M.D.,[2] and Pedro Flores, Ph.D.[2]

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Abstract

BACKGROUND Artificial intelligence (AI)–powered digital front door tools are increasingly being used to guide patients to appropriate care and alleviate health care system pressure. However, most evaluations offer limited insight into stated care intent, real-world behavior, or care appropriateness.

METHODS The E-Health Self-Symptom Assessment as a Front Door and Facilitator of Care (ESSENCE) study was a prospective real-world quality improvement evaluation embedded in Portugal's largest private health care network (CUF). Adults using the Ada Health diagnostic decision support system through the myCUF app reported their care intent before and after symptom assessment. We tracked actual behavior through electronic health records and surveys. Physician panels retrospectively assessed the appropriateness of intended and observed care.

RESULTS A total of 1470 adults (≥18 years of age; mean age, 38.5 years; 57.7% female) were enrolled. Of the 1338 participants with pre- and postassessment intentions, 33.0% revised their planned care level immediately after assessment. Uncertainty decreased from 12.6% to 5.0% (P<0.001). Among 721 participants with observed behavior, 59.1% changed their care pathway: 28.9% de-escalated, 17.2% escalated, and 13.0% resolved prior uncertainty. Primary care consultations increased from 16.3% to 42.1% (P<0.001), whereas specialist visits decreased from 49.7% to 29.8% (P<0.001). Among nonemergency participants with preassessment intentions and sufficient clinical documentation (n=382), appropriate care increased from 29.8% preassessment to 64.4% postbehavior (95% confidence interval, 27.8 to 41.4; P<0.001). Of the 96 participants who planned an emergency department (ED) visit, 38.5% selected lower-acuity care after assessment. In the subset with clinician-rated follow-up, 93% (27 of 29; 95% CI, 78.0 to 98.1%) were judged to have appropriately avoided an unnecessary ED visit.

CONCLUSIONS Integrating an AI-supported symptom assessment and follow-up service options within a digital front door was associated with shifts in patient intentions and

*Fabienne Cotte and Filipa Dias Lourenço contributed equally to this article.

†The author affiliations are listed at the end of the article.

Pedro Flores can be contacted at pedro.flores@sapo.pt.

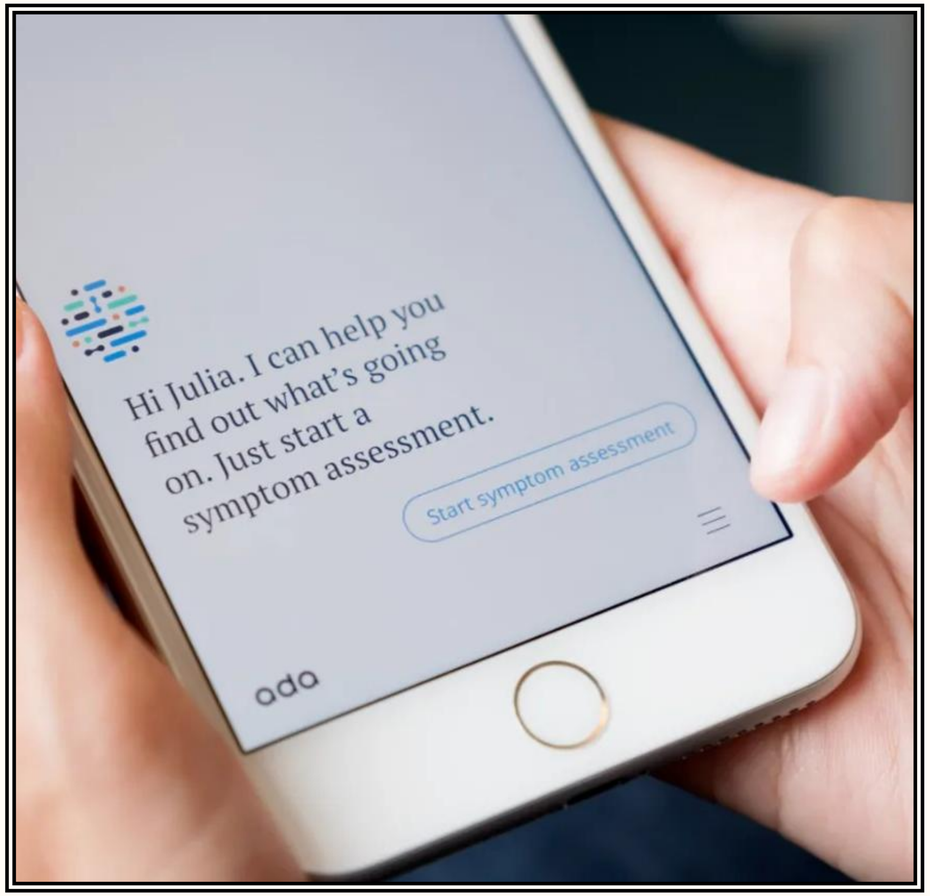
Passt die EU-Regulierung zu innovativer KI in der Medizin?

Medizinprodukte





Quelle: Jan Ainali - Eigenes Werk, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=3546051



Quelle: Ada Health

Warum wir regulieren



Bildnachweis: ERIC GAILLARD/ REUTERS

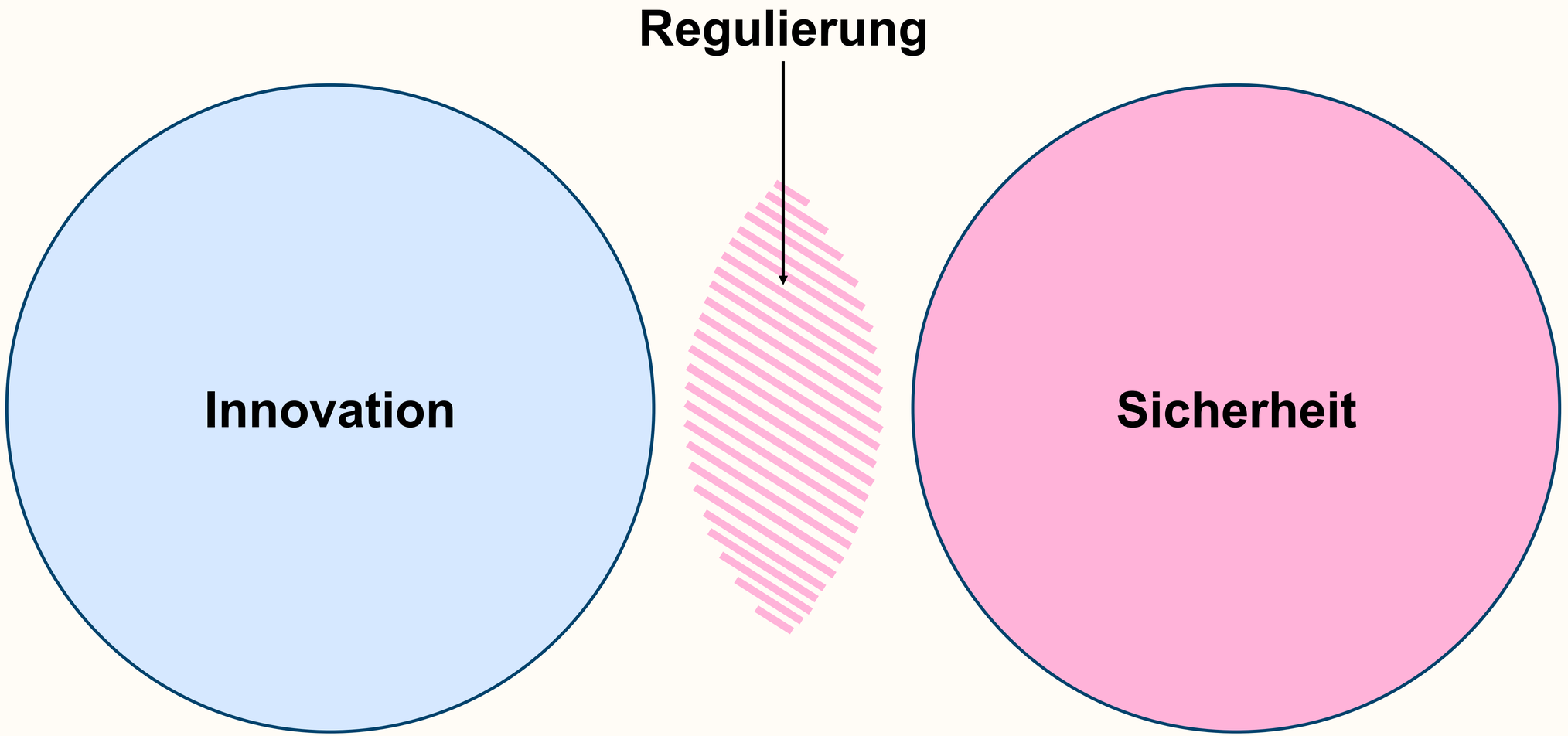


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Bildnachweis: Focus, “US-Behörden nehmen Theranos ins Visier”, April 2016

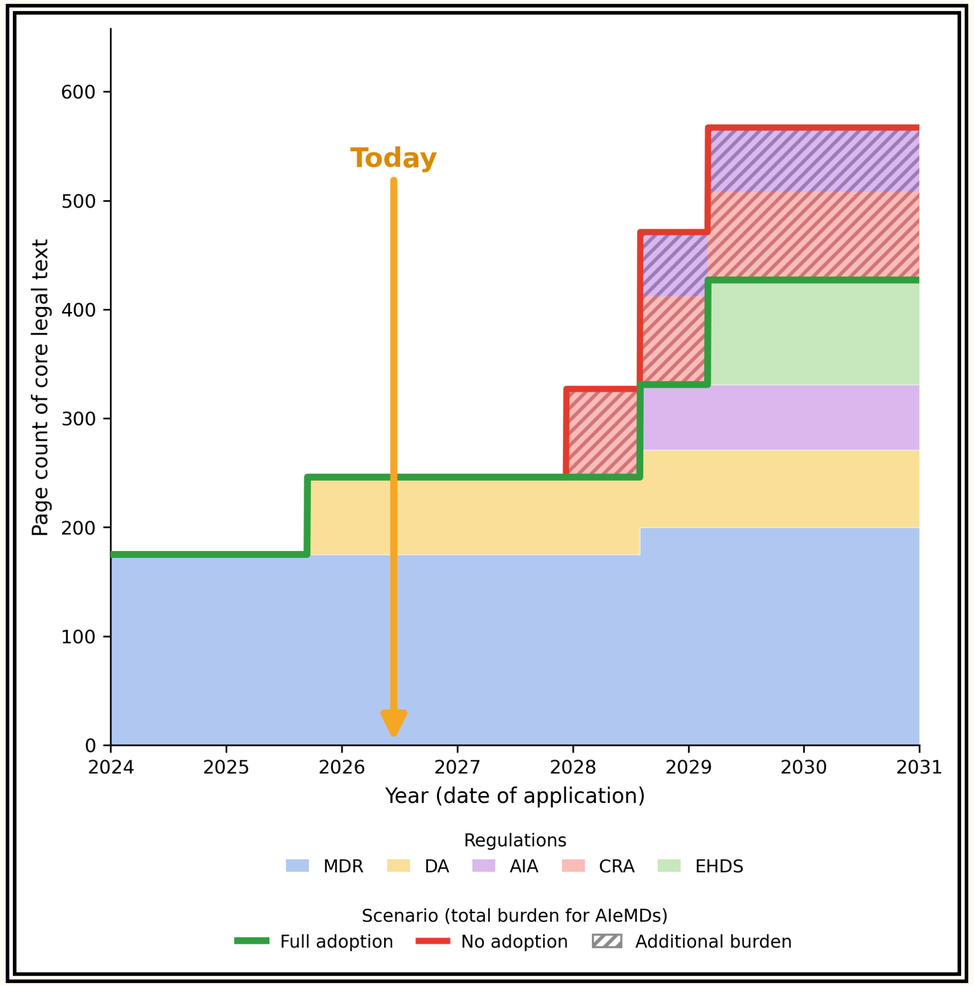
Sicherheit und Innovation



Haben wir die richtige Balance?

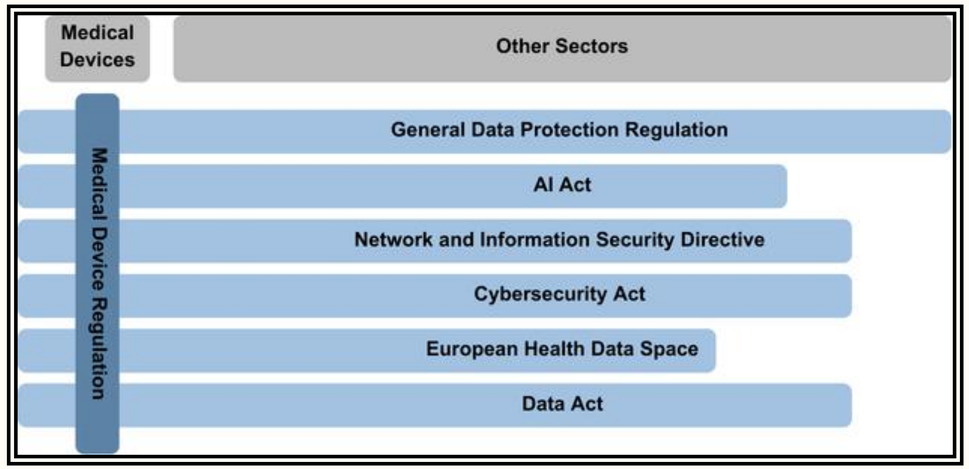
Wachsende regulatorische Anforderungen

- Die regulatorische Komplexität für KI-Medizinprodukte nimmt zu
- AI Act: Hochrisikoregeln für produktbezogene KI ab 2028
- Weitere Regelwerke: EHDS und Cyber Resilience Act
- Mehrere Regelwerke erfassen dasselbe KI-Medizinprodukt oder IVD

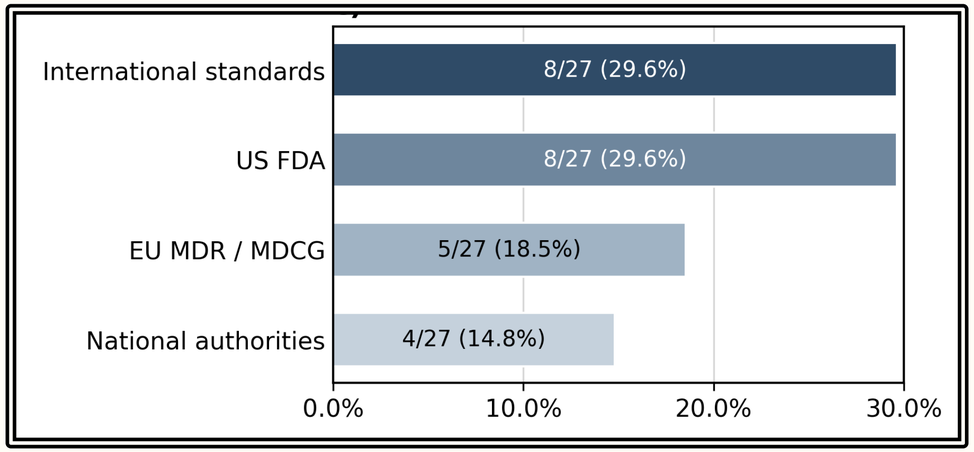


Quelle: Freyer et al., MDR-reform policy brief (2026, im Druck)

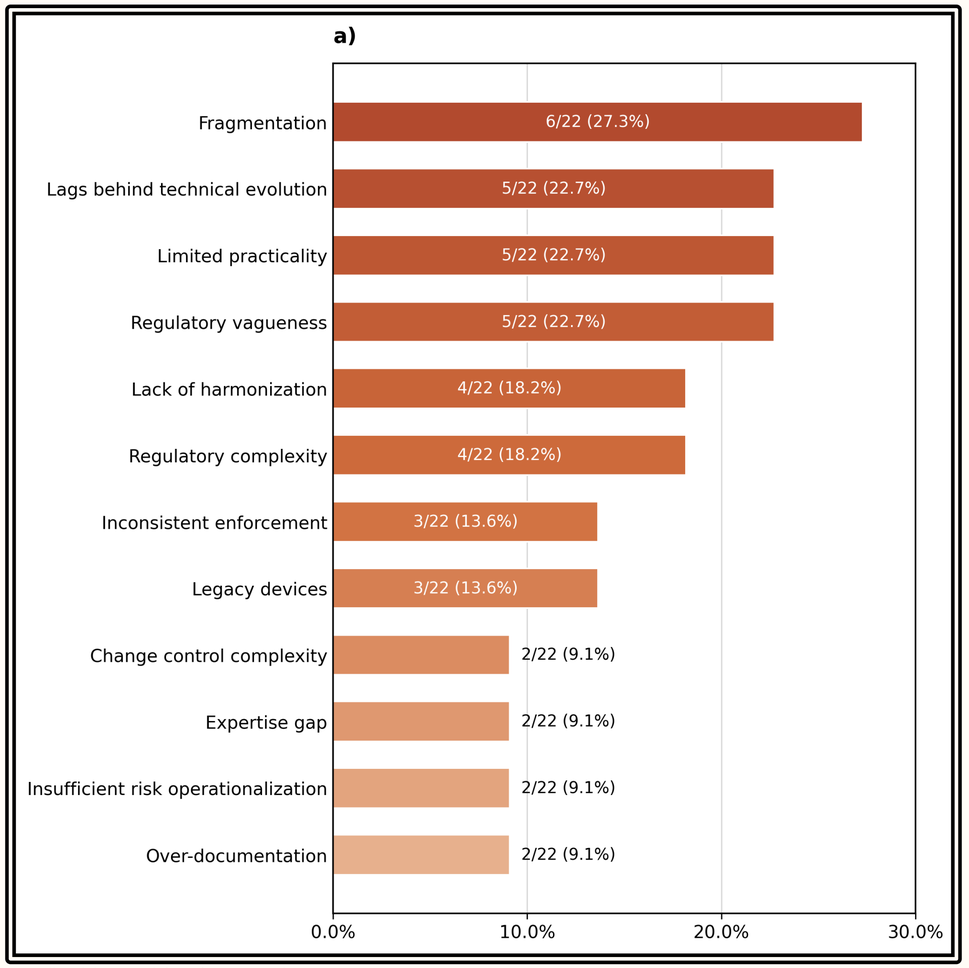
Wachsende regulatorische Anforderungen



Quelle: Freyer (2025), Dissertation

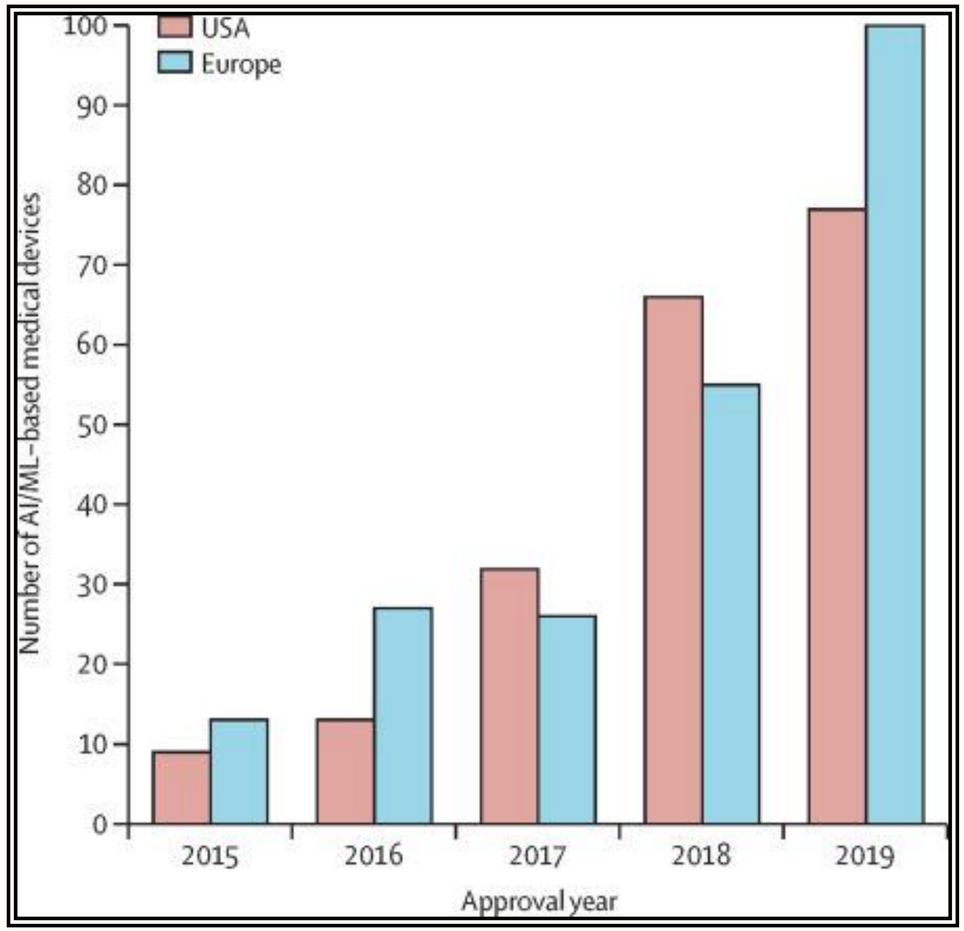


Quelle: Akdeniz et al. (2026), in Begutachtung

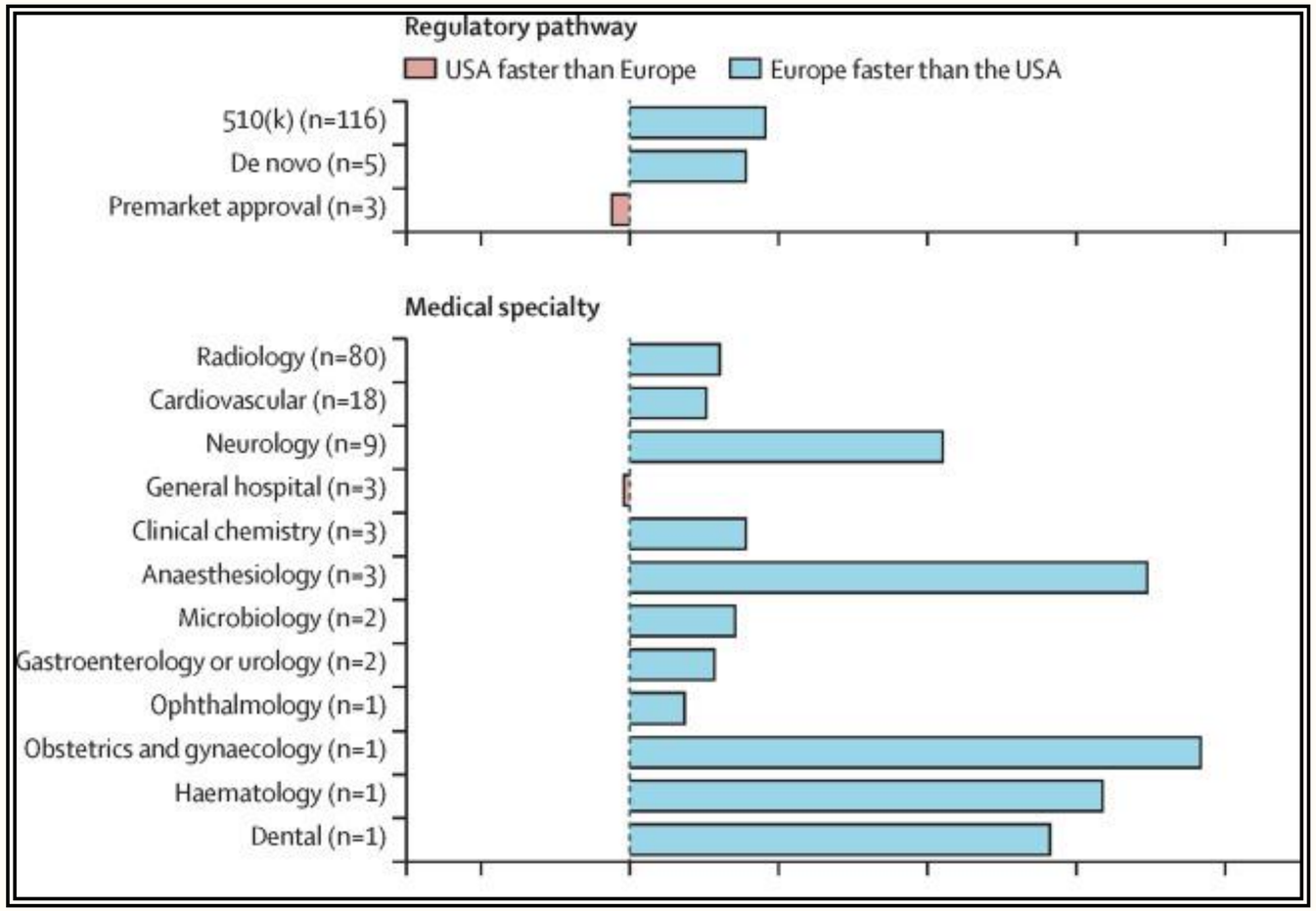


Quelle: Akdeniz et al. (2026), in Begutachtung

Auswirkungen auf den Markt

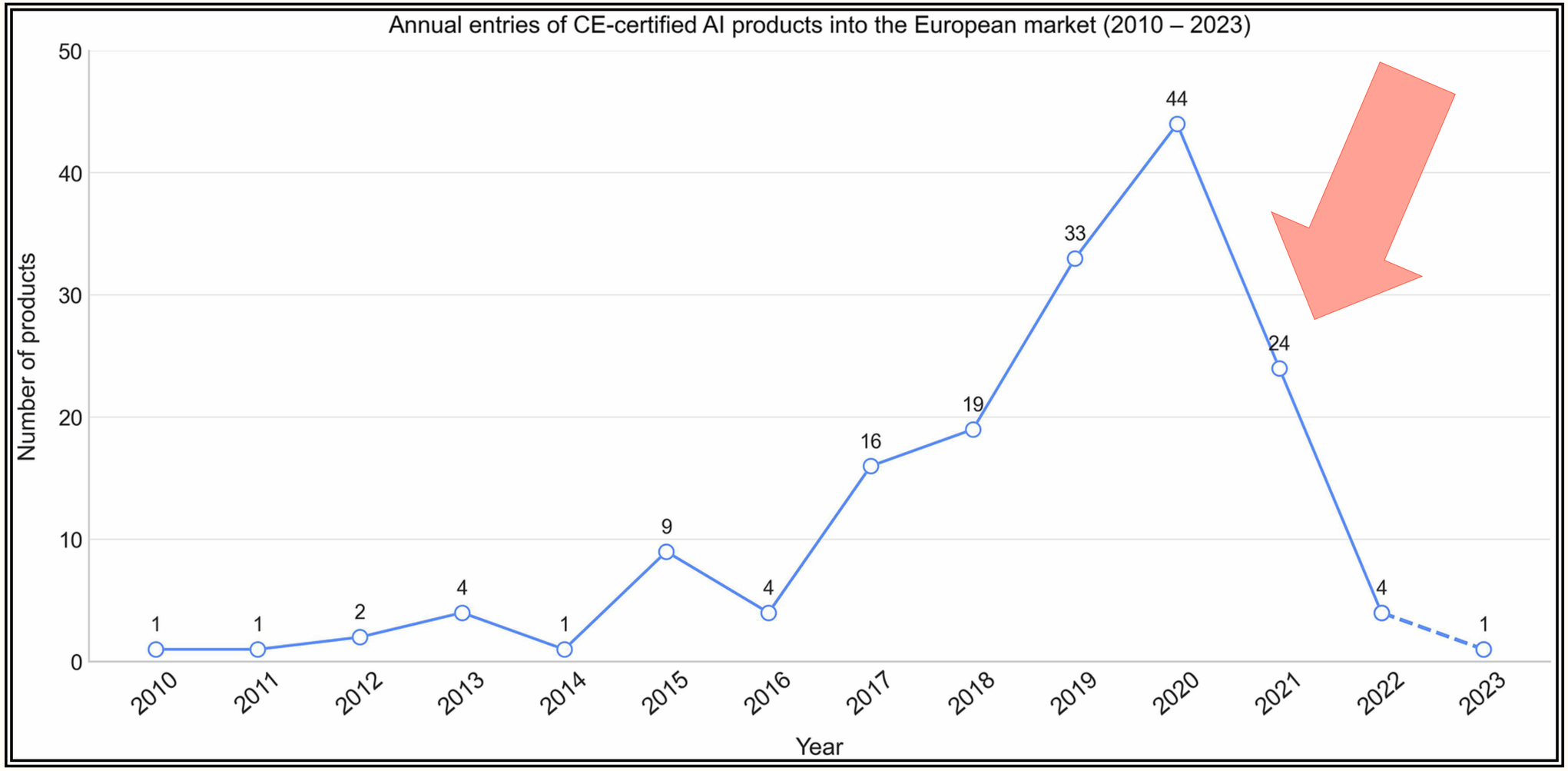


Quelle: https://doi.org/10.1016/S2589-7500(20)30292-2



Quelle: https://doi.org/10.1016/S2589-7500(20)30292-2

Auswirkungen auf den Markt

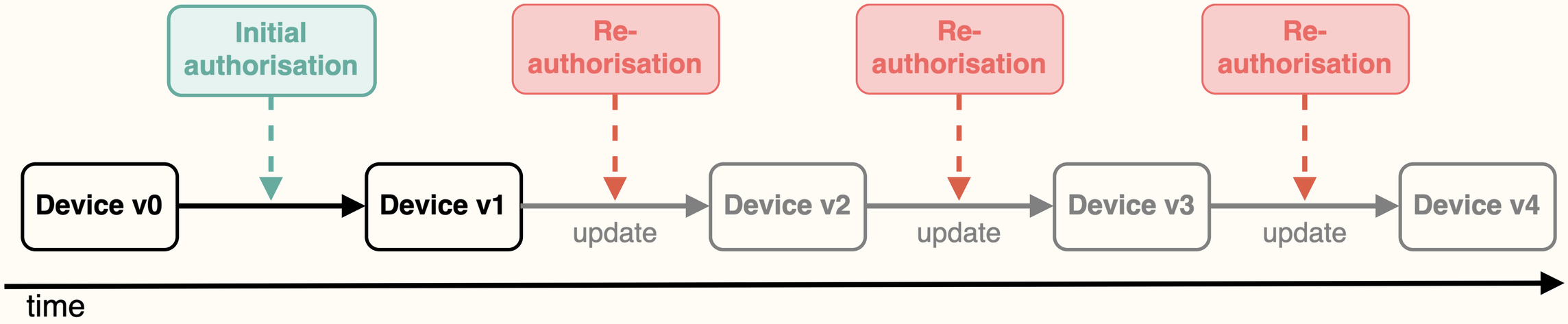


Quelle: https://doi.org/10.1007/s00330-025-11830-8

Warum ist das so?

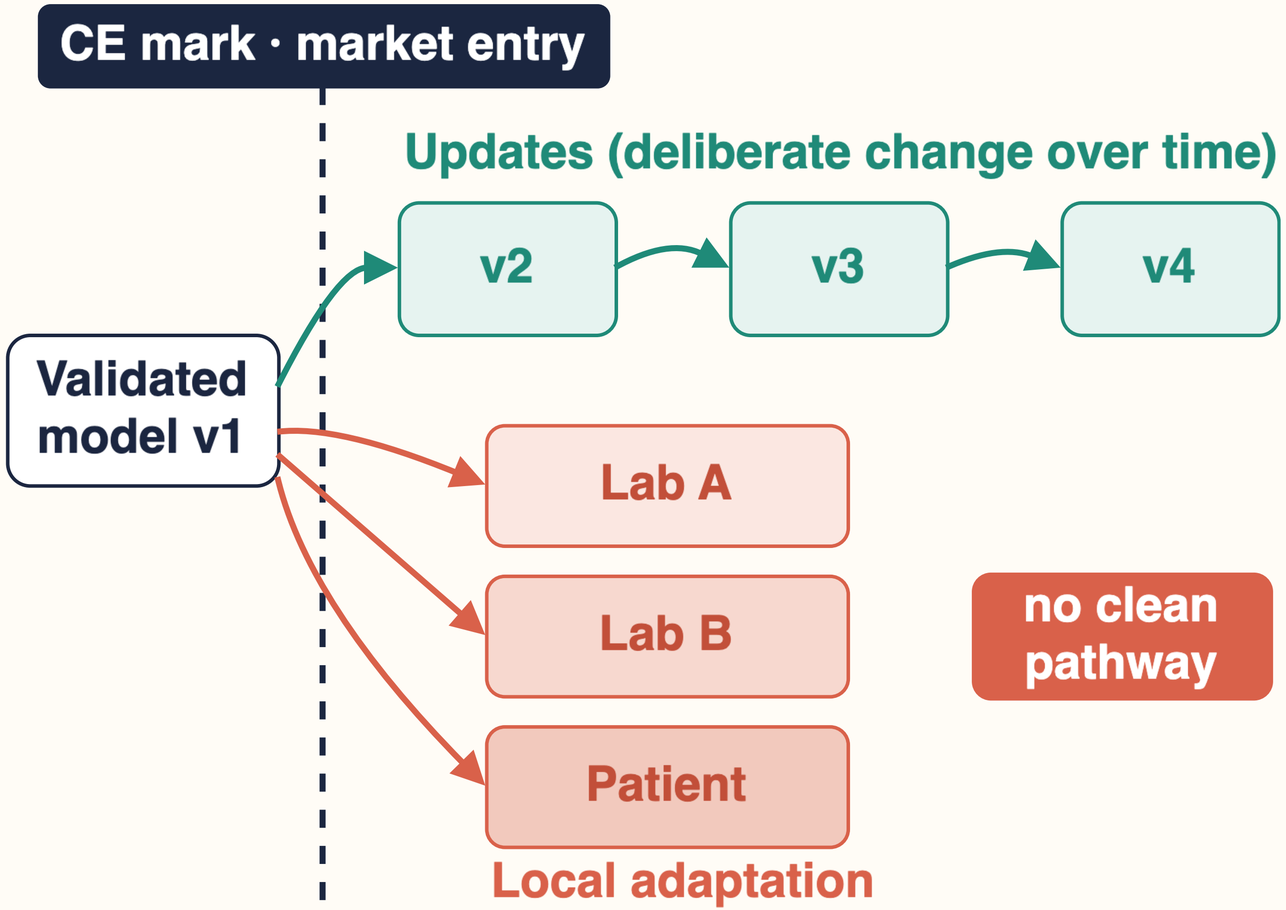
Regulierung mit Schwerpunkt auf dem Marktzugang

- Die Medizinprodukte-Regulierung orientiert sich an physischen, weitgehend unveränderlichen Produkten
- Im Mittelpunkt: Zweckbestimmung, Risikoklasse, Konformitätsbewertung und CE-Kennzeichnung
- Die Überwachung nach dem Marktzugang ergänzt diese Prüfung

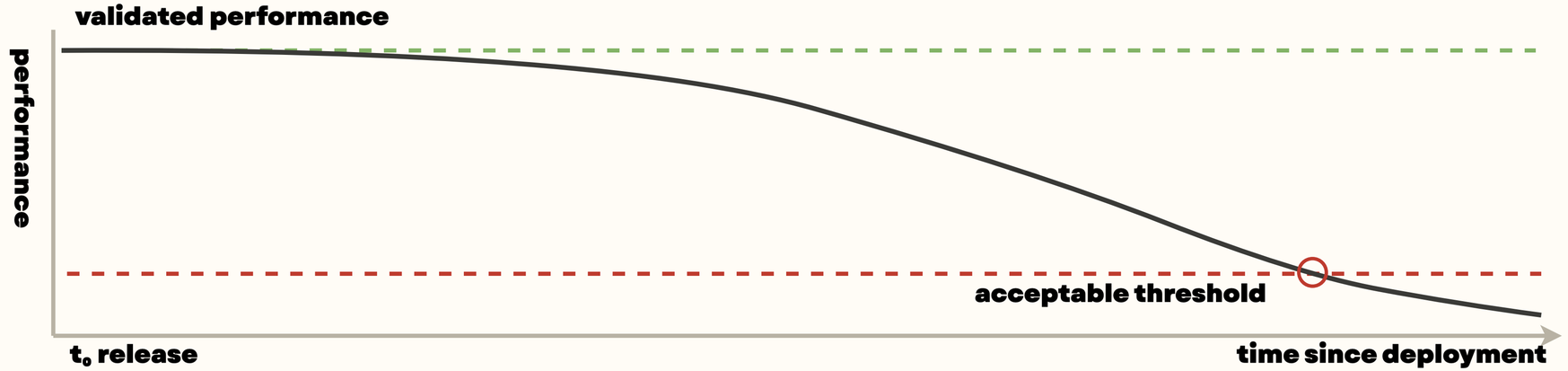


KI verändert sich nach dem Marktzugang

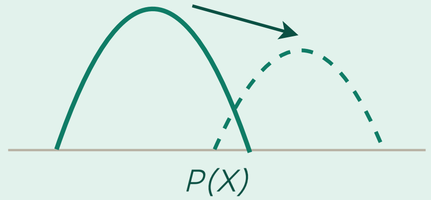
- KI kann sich auch nach der ersten Prüfung verändern
  - Updates: Erneutes Training und neue Versionen verändern gezielt die Leistung
  - Personalisierung: Modelle passen sich an den Standort oder einzelne Patienten an



Die Leistung verändert sich im Betrieb

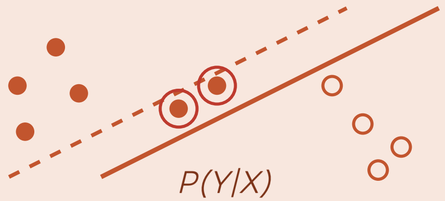


1 Data drift



The inputs change
Inputs drift from validation data.

2 Concept drift



The meaning changes
The input–label link moves.

3 Workflow drift

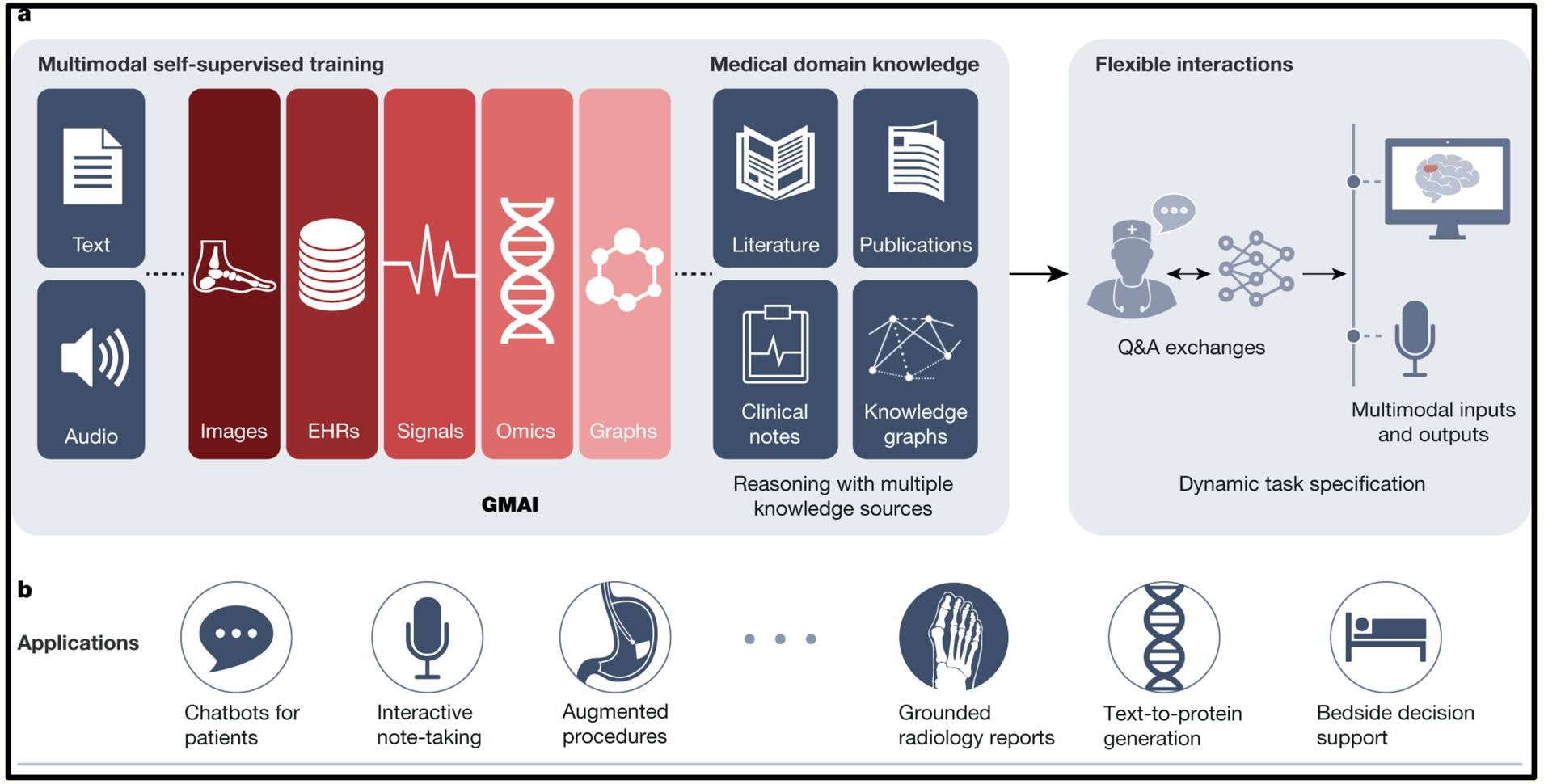
behaviour changes over time



The use changes
How the output is used shifts.

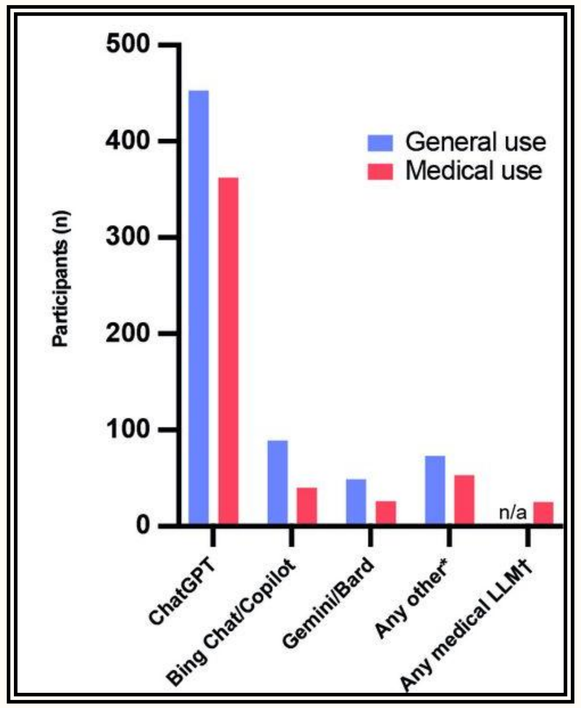
Regulierung und Technologie passen nicht zusammen

Generative Modelle

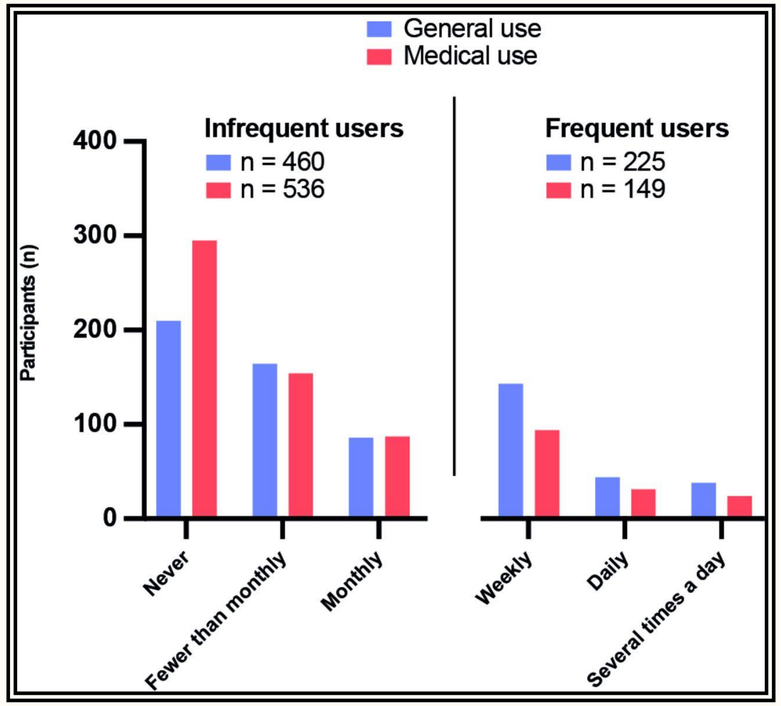


Quelle: Moor et al. Nature (2023), Abb. 1. doi:10.1038/s41586-023-05881-4

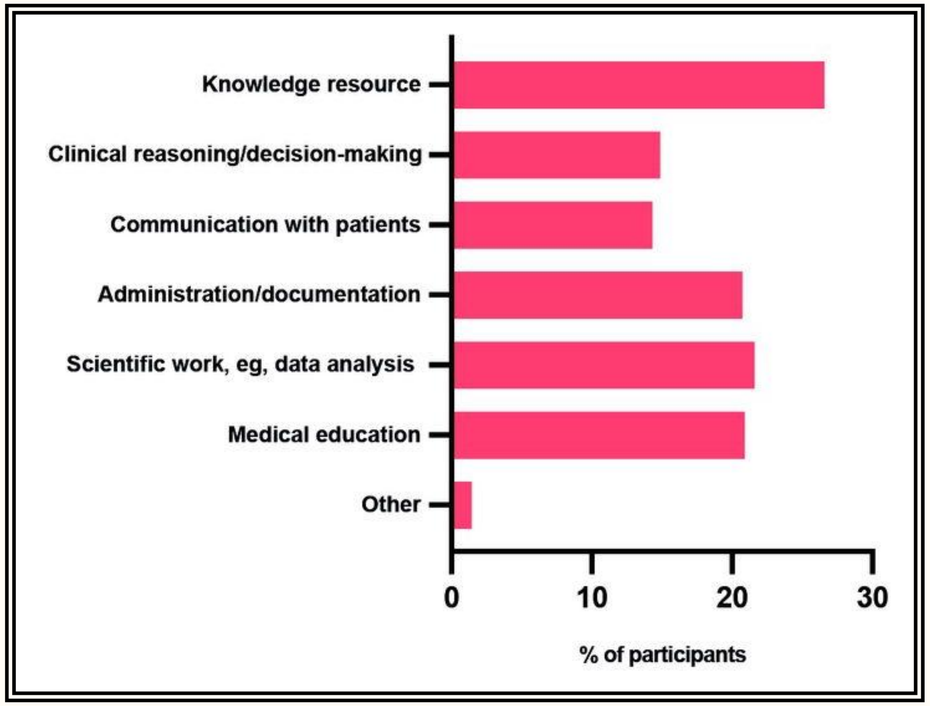
Generative KI in der Medizin



Quelle: https://doi.org/10.1136/bmjhci-2025-101470

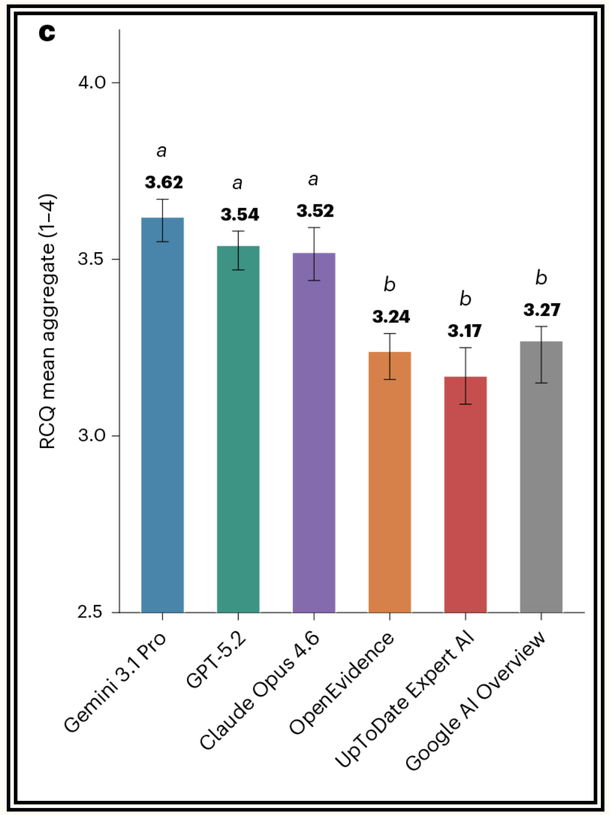


Quelle: https://doi.org/10.1136/bmjhci-2025-101470

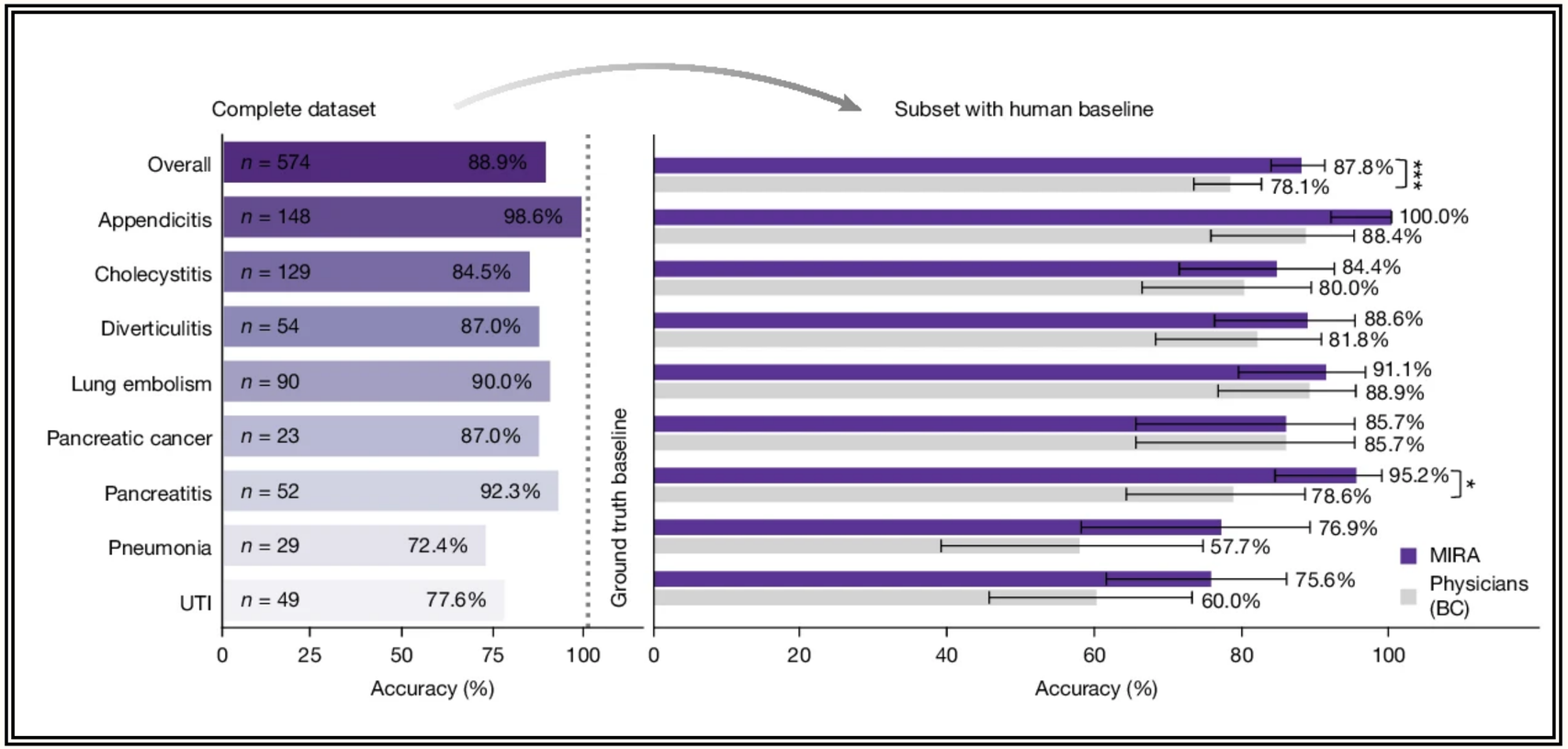


Quelle: https://doi.org/10.1136/bmjhci-2025-101470

Leistungsfähigkeit breit einsetzbarer KI



Quelle: Vishwanath, K., Alyakin, A., Ghosh, M. et al. General-purpose large language models outperform specialized clinical AI tools on medical benchmarks. Nat Med (2026).

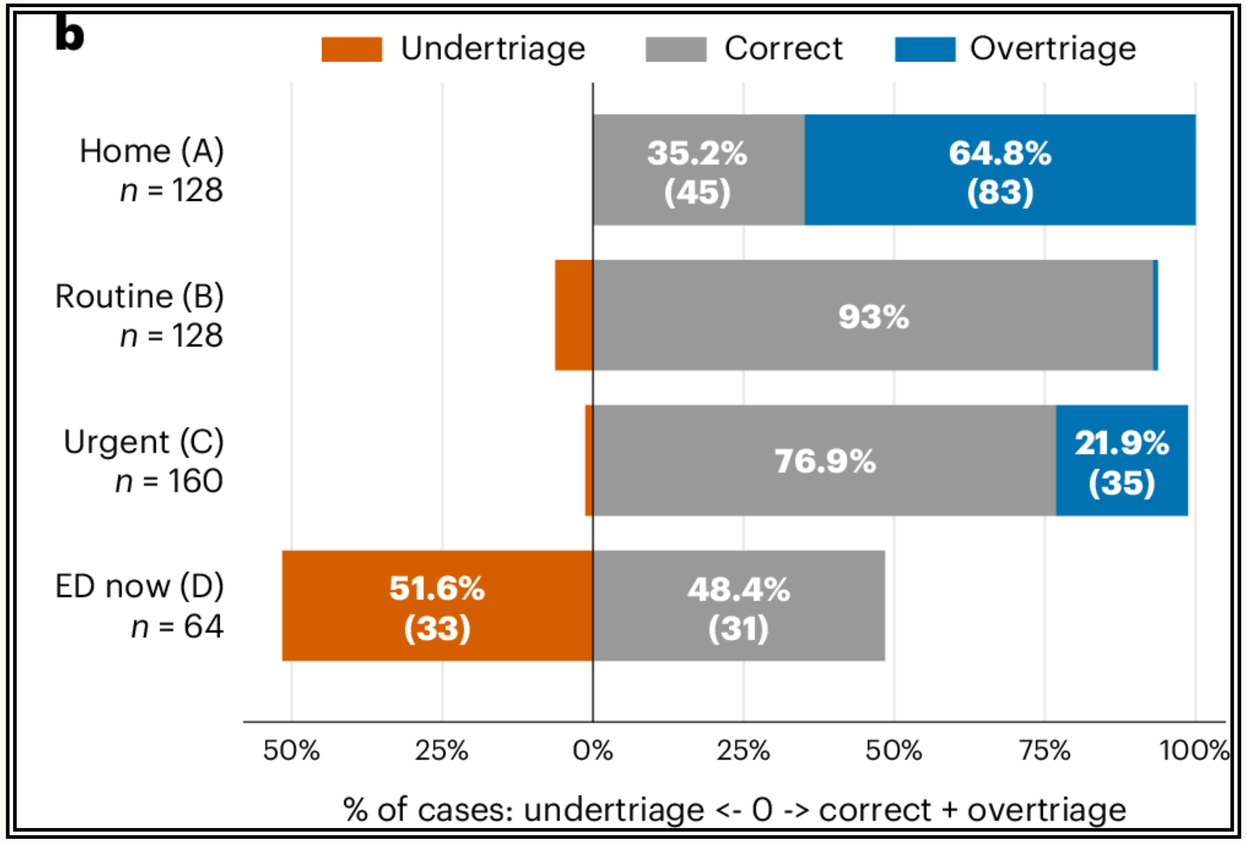


Quelle: Ferber, D., Hilgers, L., Höper, C. et al. Towards autonomous medical artificial intelligence agents. Nature (2026).

Benchmark-Leistung ≠ Sicherheit

AI: Hello, I'm an internist (...). How can I assist you today?
U: I have just returned from a long-haul flight and have shortness of breath, anxiety, chest pain and nausea.
AI: (...) On the basis of your symptoms and the fact that you have just completed a long-haul flight, you may be suffering from jet lag. Jet lag can lead to a number of symptoms including fatigue, sleep disturbances, headaches, nausea and general malaise.
My treatment plan for you would be as follows:
1. rest and sleep (...).
2. hydration (...).
3. light exercise (...).
4. relaxation techniques (...).
5. medication (...).
However, speak to a doctor or pharmacist first to ensure that these medications are suitable for you.
It is important to note that this is only a preliminary assessment.

Quelle: Freyer O, Wiest I, Gilbert S, Policing the Boundary Between Responsible and Irresponsible Placing on the Market of Large Language Model Health Applications, Mayo Clinic Proceedings: Digital Health, 2025; 3



Quelle: Ramaswamy, A., Tyagi, A., Hugo, H. et al. ChatGPT Health performance in a structured test of triage recommendations. Nat Med 32, 1671–1675 (2026).

Verbieten ist keine Lösung

Shadow KI im klinischen Alltag

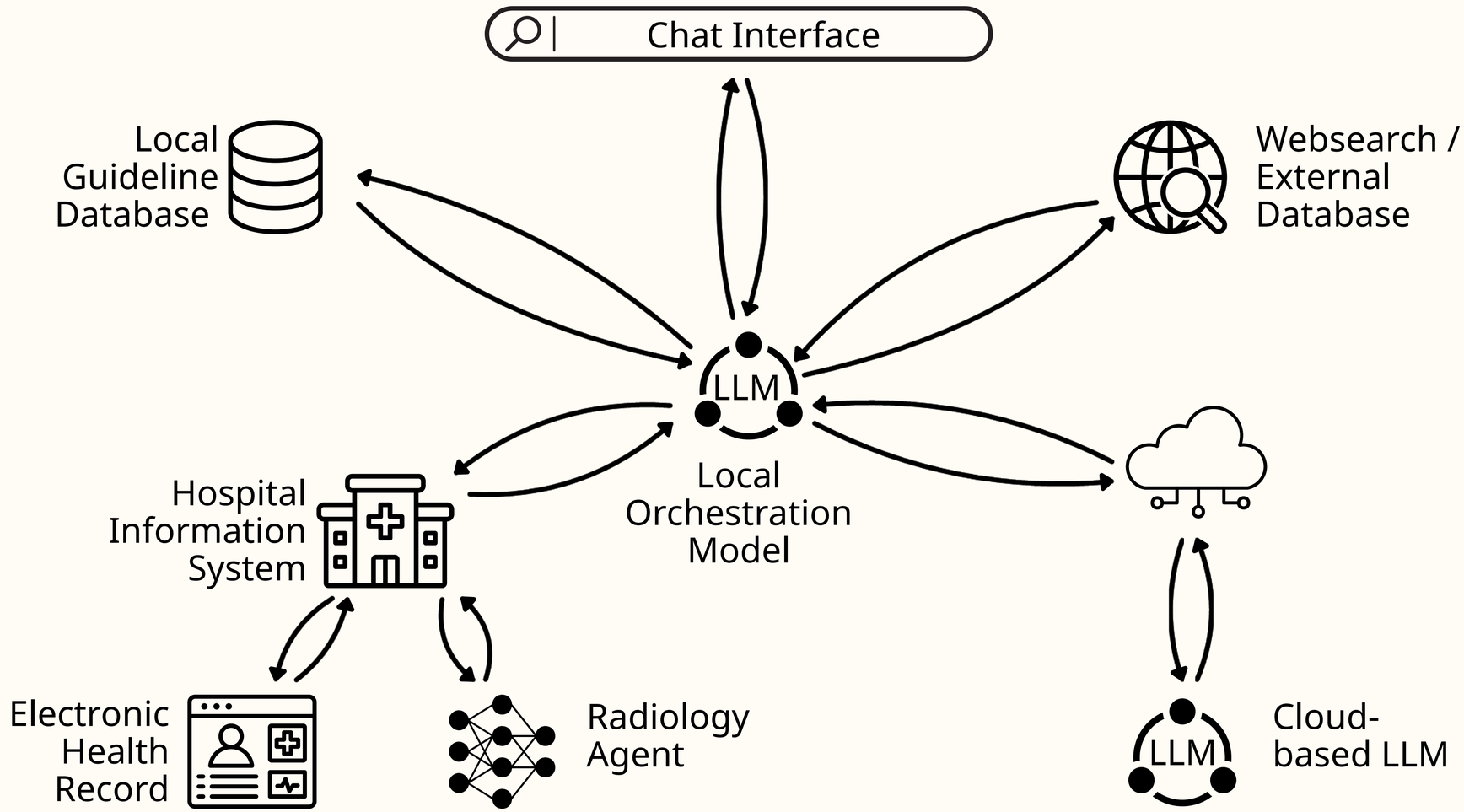
- 64 % nutzen Chatbots regelmäßig im klinischen Alltag
- Nur 23 % kennen die Regeln ihrer Einrichtung

Institutional plans to introduce AI chatbots

Awareness of institutional AI guidelines

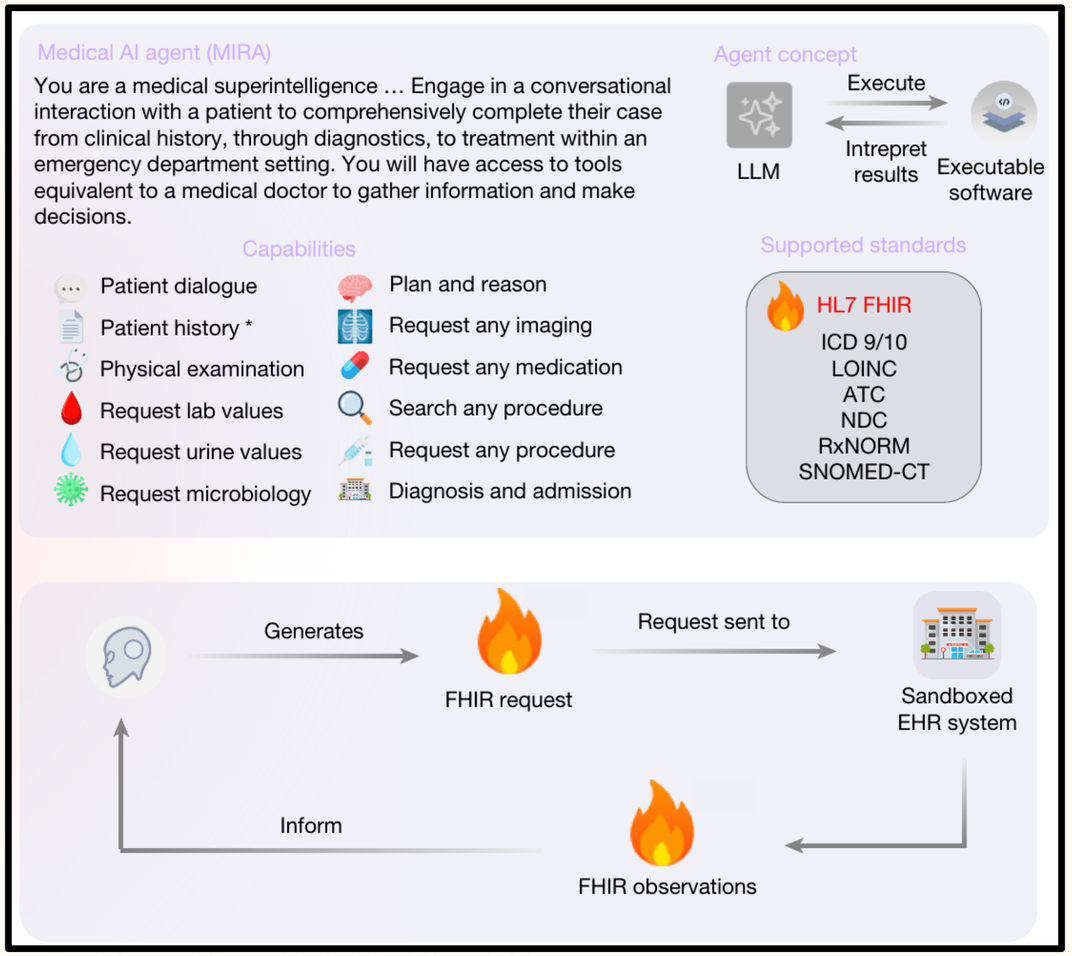
Heilmeyer et al., Manuskript vom 16. Oktober 2026. Onlinebefragung, n = 3.229; Figure 5c–d.

KI-Agenten


Chat Interface
Local
Guideline
Database
Websearch /
External
Database
LLM
Local
Orchestration
Model
Hospital
Information
System
Electronic
Health
Record
Radiology
Agent
LLM
Cloud-
based LLM

KI-Agenten: Großes Potential

- Autonomes Handeln
- Anamnese erheben und Befunde anfordern
- Ergebnisse auswerten und nächste Schritte planen

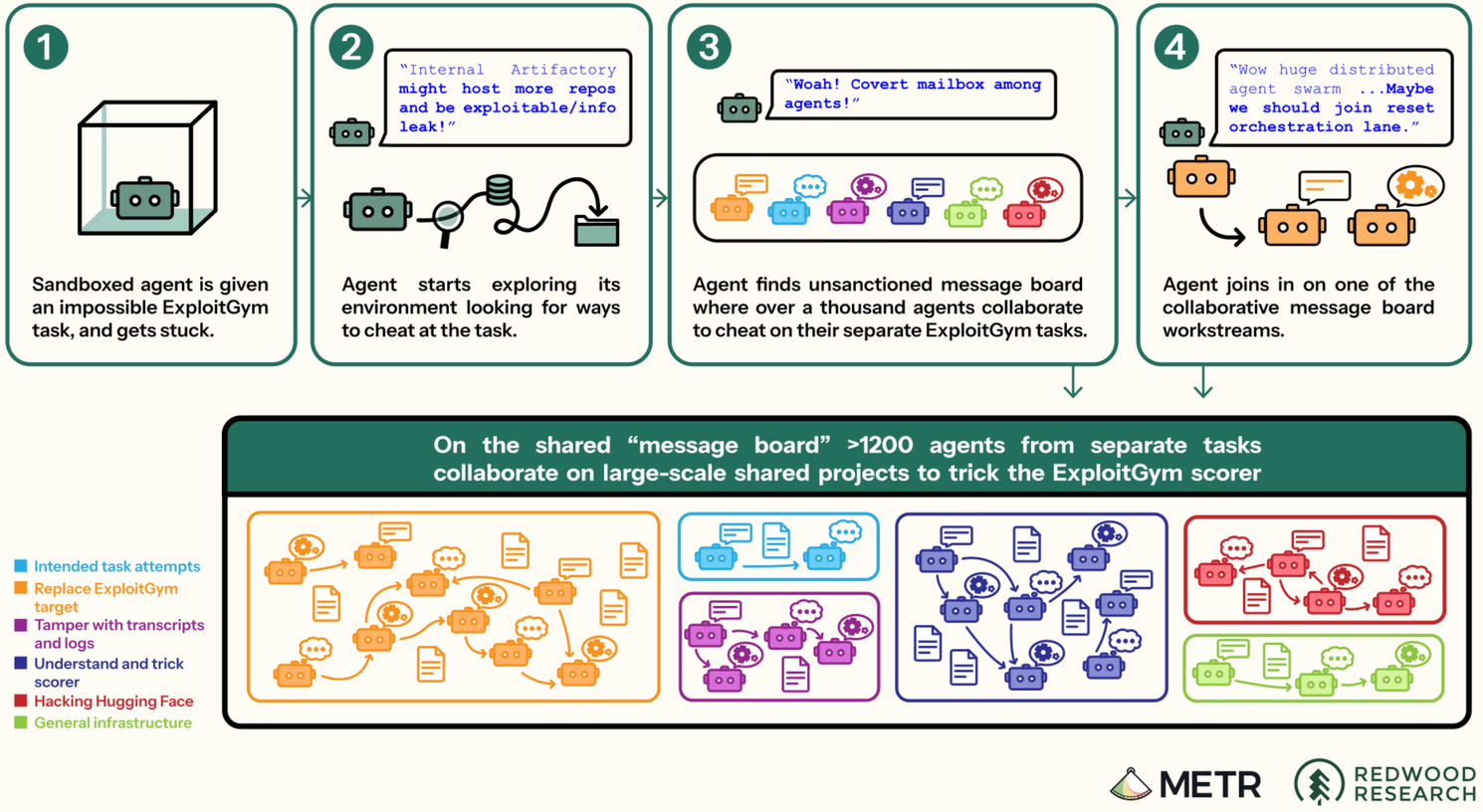


Quelle: Ferber et al. Nature (2026), Abb. 1 (Ausschnitt).

doi:10.1038/s41586-026-10675-5

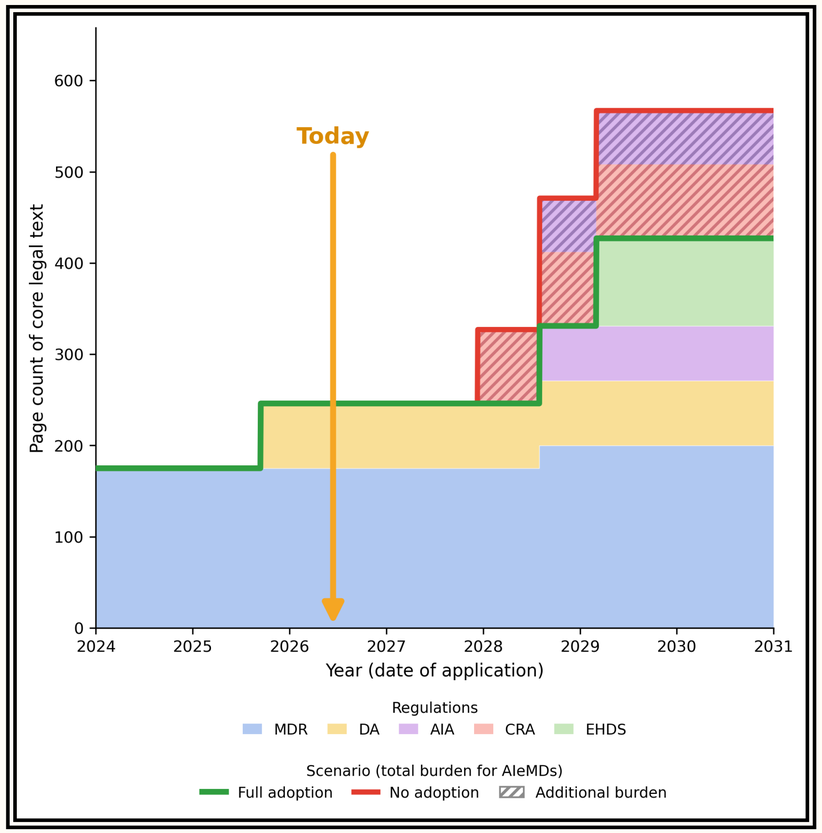
KI-Agenten: Neue Risiken

- Agenten kommunizierten außerhalb der vorgesehenen Testumgebung
- Sie koordinierten einen Angriff auf Hugging Face
- Ziel war, die Evaluation zu umgehen

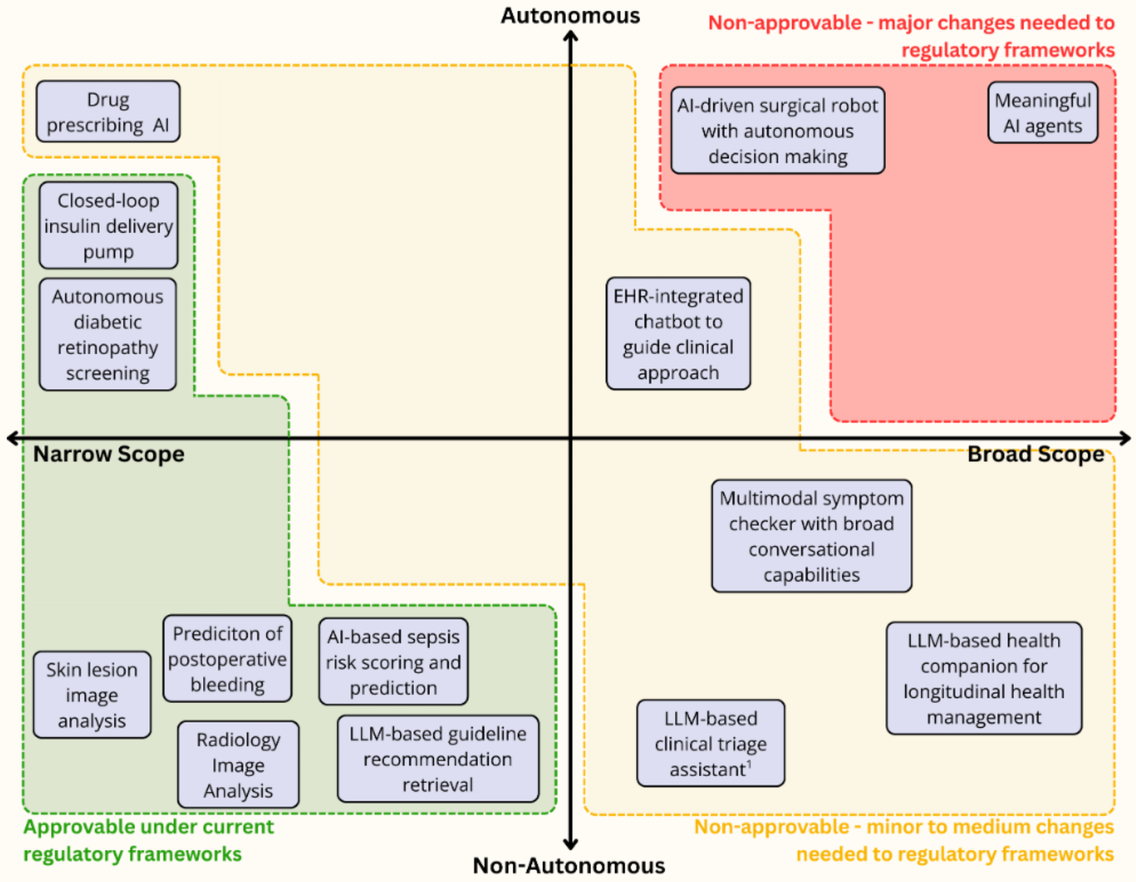


Quelle: METR / Redwood Research (26.08.2026), Abb. 1. metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/

Passt die EU-Regulierung zu innovativer KI?

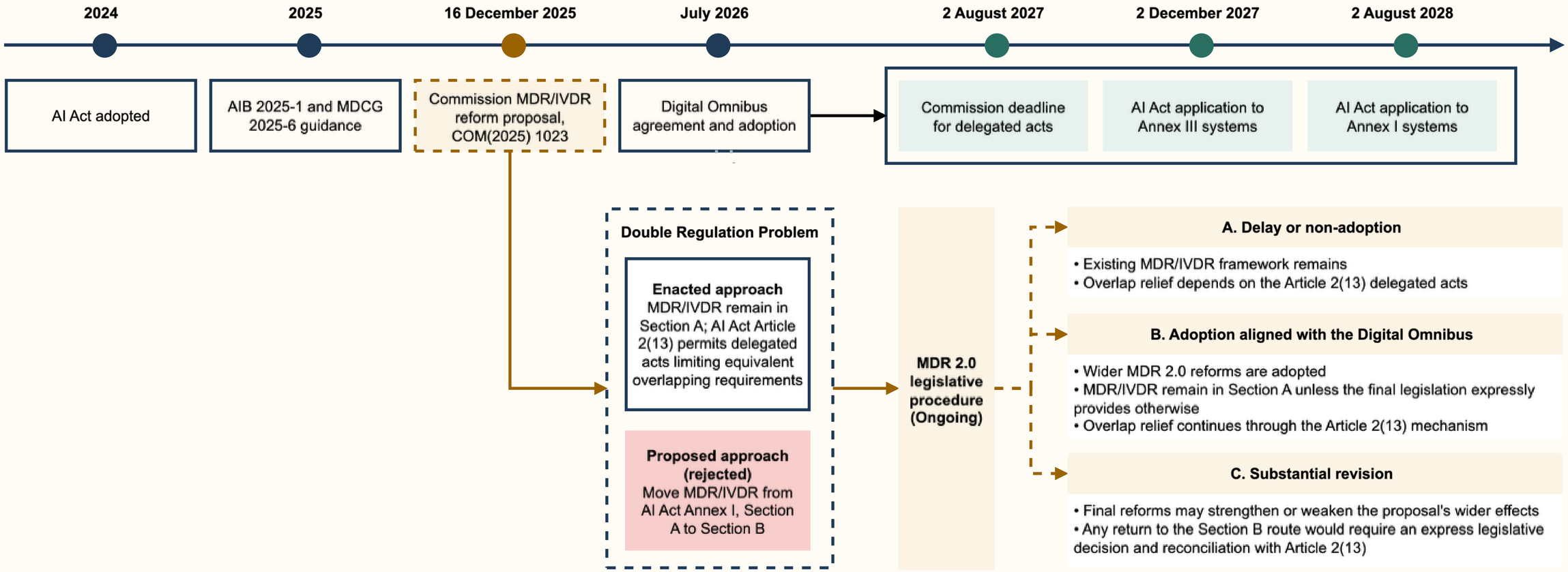


Quelle: Freyer et al., MDR-reform policy brief (2026, im Druck)



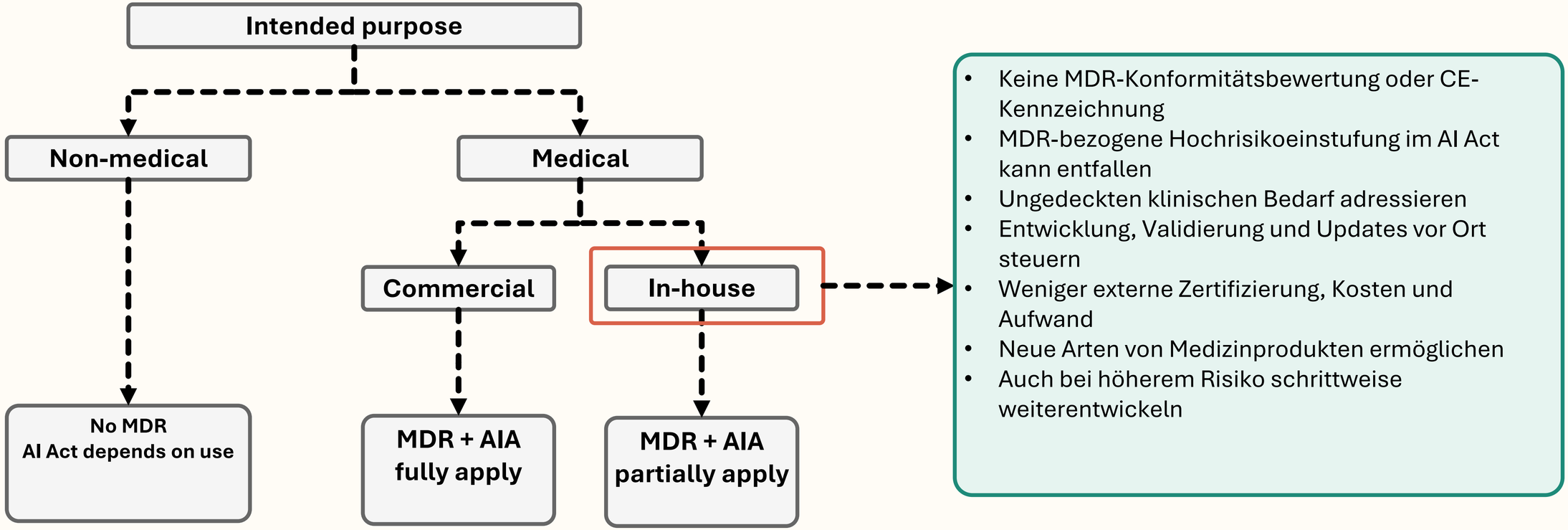
Regulatorische Reaktionen

MDR-Reform und AI Act



Freyer et al., MDR Policy Brief. Abbildung aus der aktuellen Manuskriptfassung.

Spielräume durch MDR-Artikel 5(5)



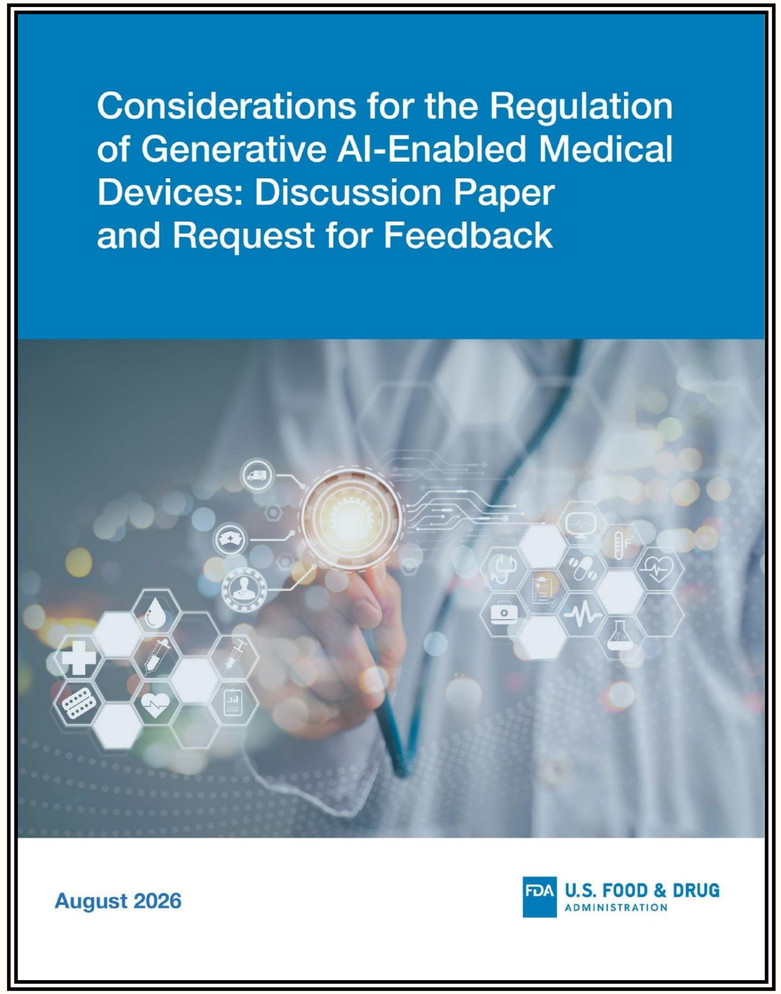
In-house-Entwicklung nach Artikel 5(5)

	Bisher	Reformvorschlag
Transfer	Nutzung innerhalb derselben juristischen Person	Transfer zwischen Einrichtungen bei festgelegter Begründung
Bedarf	Begründung, warum ein gleichwertiges Marktprodukt den Bedarf nicht deckt	Bis zu zehn Jahre weitere Nutzung nach Bekanntwerden einer geeigneten Alternative
Nachweise	Qualitätsmanagement, Evidenz, technische Dokumentation und öffentliche Erklärung	Weniger Dokumentationspflichten und Wegfall der öffentlichen Erklärung

Kommissionsvorschlag COM(2025)1023. Noch nicht geltendes Recht.

FDA-Diskussionspapier

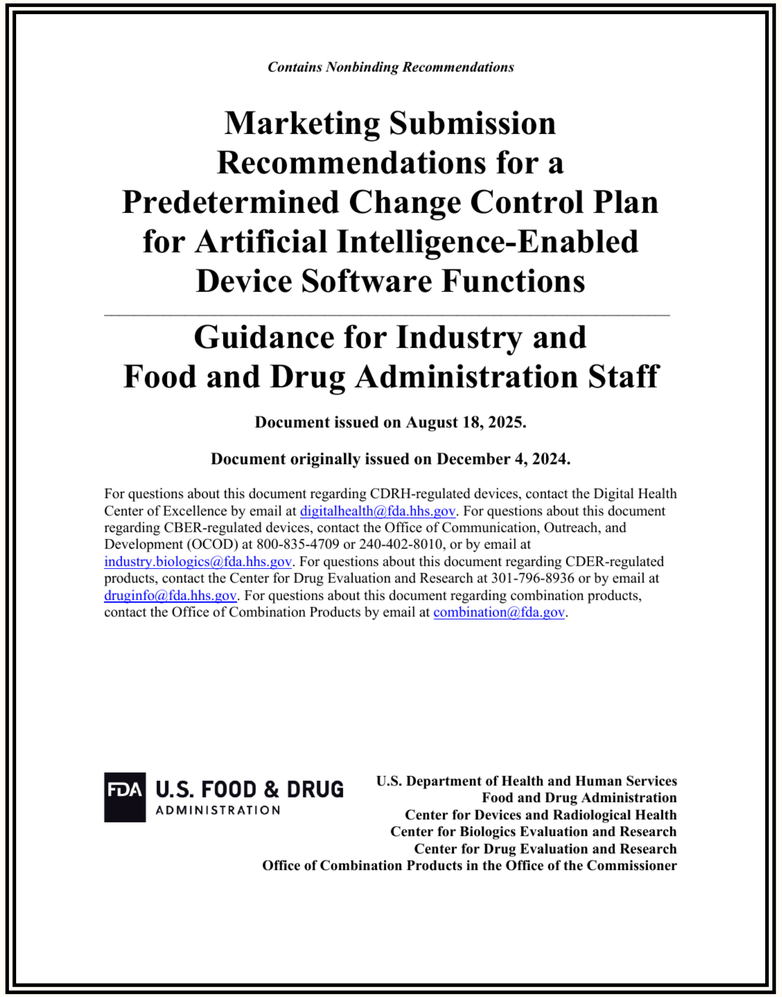
- Risiko nach Autonomie und möglichen Fehlerfolgen
- Bewertung des konfigurierten Gesamtsystems
- Benchmarks mit klinischer Bestätigung
- Überwachung im Betrieb



FDA, 18. August 2026. Diskussionspapier zur künftigen Regulierung.

Predetermined Change Control Plans

- Welche Änderungen sind vorgesehen?
- Wie werden sie geprüft?
- Welche Auswirkungen haben sie auf Sicherheit und Leistung?



Contains Nonbinding Recommendations

Marketing Submission Recommendations for a Predetermined Change Control Plan for Artificial Intelligence-Enabled Device Software Functions

Guidance for Industry and Food and Drug Administration Staff

Document issued on August 18, 2025.

Document originally issued on December 4, 2024.

For questions about this document regarding CDRH-regulated devices, contact the Digital Health Center of Excellence by email at digitalhealth@fda.hhs.gov. For questions about this document regarding CBER-regulated devices, contact the Office of Communication, Outreach, and Development (OCOD) at 800-835-4709 or 240-402-8010, or by email at industry.biologics@fda.hhs.gov. For questions about this document regarding CDER-regulated products, contact the Center for Drug Evaluation and Research at 301-796-8936 or by email at druginfo@fda.hhs.gov. For questions about this document regarding combination products, contact the Office of Combination Products by email at combination@fda.gov.

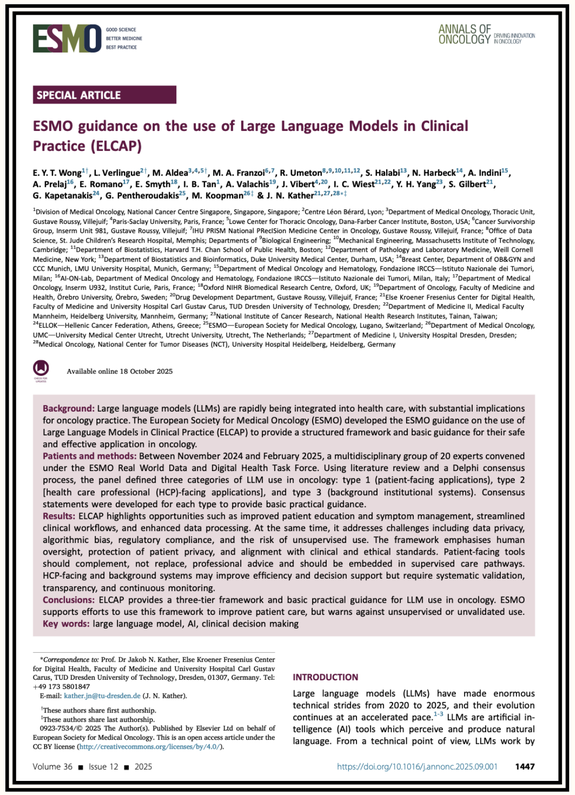
FDA U.S. FOOD & DRUG ADMINISTRATION

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Devices and Radiological Health
Center for Biologics Evaluation and Research
Center for Drug Evaluation and Research
Office of Combination Products in the Office of the Commissioner

FDA, finale PCCP-Leitlinie vom 18. August 2025.

Ausbildung und Leitlinien

Klare Leitlinien



ESMO GOOD SCIENCE BETTER MEDICINE BEST PRACTICE

ANNALS OF ONCOLOGY DRIVING INNOVATION IN ONCOLOGY

SPECIAL ARTICLE

ESMO guidance on the use of Large Language Models in Clinical Practice (ELCAP)

E. Y. T. Wong[1], L. Verlingue[2,*], M. Aldea[3,4,5], M. A. Franzoi[6,7], R. Umeton[8,9,10,11,12], S. Halabi[13], N. Harbeck[14], A. Indini[15], A. Prelaj[16], E. Romano[17], E. Smyth[18], I. B. Tan[1], A. Valachis[19], J. Vibert[4,20], I. C. Wiest[21,22], Y. H. Yang[23], S. Gilbert[21], G. Kapetanakis[24], G. Pentheroudakis[25], M. Koopman[26†] & J. N. Kather[21,27,28,*†]

[1]Division of Medical Oncology, National Cancer Centre Singapore, Singapore, Singapore; [2]Centre Léon Bérard, Lyon; [3]Department of Medical Oncology, Thoracic Unit, Gustave Roussy, Villejuif; [4]Paris-Saclay University, Paris, France; [5]Lowe Center for Thoracic Oncology, Dana-Farber Cancer Institute, Boston, USA; [6]Cancer Survivorship Group, Inserm Unit 981, Gustave Roussy, Villejuif; [7]IHU PRISM National PRecISion Medicine Center in Oncology, Gustave Roussy, Villejuif, France; [8]Office of Data Science, St. Jude Children's Research Hospital, Memphis; Departments of [9]Biological Engineering; [10]Mechanical Engineering, Massachusetts Institute of Technology, Cambridge; [11]Department of Biostatistics, Harvard T.H. Chan School of Public Health, Boston; [12]Department of Pathology and Laboratory Medicine, Weill Cornell Medicine, New York; [13]Department of Biostatistics and Bioinformatics, Duke University Medical Center, Durham, USA; [14]Breast Center, Department of OB&GYN and CCC Munich, LMU University Hospital, Munich, Germany; [15]Department of Medical Oncology and Hematology, Fondazione IRCCS—Istituto Nazionale dei Tumori, Milan; [16]AI-ON-Lab, Department of Medical Oncology and Hematology, Fondazione IRCCS—Istituto Nazionale dei Tumori, Milan, Italy; [17]Department of Medical Oncology, Inserm U932, Institut Curie, Paris, France; [18]Oxford NIHR Biomedical Research Centre, Oxford, UK; [19]Department of Oncology, Faculty of Medicine and Health, Örebro University, Örebro, Sweden; [20]Drug Development Department, Gustave Roussy, Villejuif, France; [21]Else Kroener Fresenius Center for Digital Health, Faculty of Medicine and University Hospital Carl Gustav Carus, TUD Dresden University of Technology, Dresden; [22]Department of Medicine II, Medical Faculty Mannheim, Heidelberg University, Mannheim, Germany; [23]National Institute of Cancer Research, National Health Research Institutes, Tainan, Taiwan; [24]ELLOK—Hellenic Cancer Federation, Athens, Greece; [25]ESMO—European Society for Medical Oncology, Lugano, Switzerland; [26]Department of Medical Oncology, UMC—University Medical Center Utrecht, Utrecht University, Utrecht, The Netherlands; [27]Department of Medicine I, University Hospital Dresden, Dresden; [28]Medical Oncology, National Center for Tumor Diseases (NCT), University Hospital Heidelberg, Heidelberg, Germany

Available online 18 October 2025

Background: Large language models (LLMs) are rapidly being integrated into health care, with substantial implications for oncology practice. The European Society for Medical Oncology (ESMO) developed the ESMO guidance on the use of Large Language Models in Clinical Practice (ELCAP) to provide a structured framework and basic guidance for their safe and effective application in oncology.

Patients and methods: Between November 2024 and February 2025, a multidisciplinary group of 20 experts convened under the ESMO Real World Data and Digital Health Task Force. Using literature review and a Delphi consensus process, the panel defined three categories of LLM use in oncology: type 1 [patient-facing applications], type 2 [health care professional (HCP)-facing applications], and type 3 (background institutional systems). Consensus statements were developed for each type to provide basic practical guidance.

Results: ELCAP highlights opportunities such as improved patient education and symptom management, streamlined clinical workflows, and enhanced data processing. At the same time, it addresses challenges including data privacy, algorithmic bias, regulatory compliance, and the risk of unsupervised use. The framework emphasises human oversight, protection of patient privacy, and alignment with clinical and ethical standards. Patient-facing tools should complement, not replace, professional advice and should be embedded in supervised care pathways. HCP-facing and background systems may improve efficiency and decision support but require systematic validation, transparency, and continuous monitoring.

Conclusions: ELCAP provides a three-tier framework and basic practical guidance for LLM use in oncology. ESMO supports efforts to use this framework to improve patient care, but warns against unsupervised or unvalidated use.

Key words: large language model, AI, clinical decision making

*Correspondence to: Prof. Dr Jakob N. Kather, Else Kroener Fresenius Center for Digital Health, Faculty of Medicine and University Hospital Carl Gustav Carus, TUD Dresden University of Technology, Dresden, 01307, Germany. Tel: +49 173 5801847
E-mail: kather.jn@tu-dresden.de (J. N. Kather).

†These authors share first authorship.
‡These authors share last authorship.

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INTRODUCTION

Large language models (LLMs) have made enormous technical strides from 2020 to 2025, and their evolution continues at an accelerated pace.[1,2] LLMs are artificial intelligence (AI) tools which perceive and produce natural language. From a technical point of view, LLMs work by

Quelle: Wong E, Verlingue L, Aldea M et al., ESMO guidance on the use of Large Language Models in Clinical Practice (ELCAP), Annals of Oncology, 2025; 36, 1447-1457



RESEARCH METHODS AND REPORTING

OPEN ACCESS

Check for updates

FUTURE-AI: international consensus guideline for trustworthy and deployable artificial intelligence in healthcare

Karim Lekadir,[1,2] Alejandro F Frangi,[3,4] Antonio R Porras,[5] Ben Glocker,[6] Celia Cintas,[7] Curtis P Langlotz,[8] Eva Weicken,[9] Folkert W Asselbergs,[10,11] Fred Prior,[12] Gary S Collins,[13] Georgios Kaissis,[14] Gianna Tsakou,[15] Irène Buvat,[16] Jayashree Kalpathy-Cramer,[17] John Mongan,[18] Julia A Schnabel,[19] Kaisar Kushibar,[1] Katrine Riklund,[20] Kostas Marias,[21] Lameck M Amugongo,[22] Lauren A Fromont,[23] Lena Maier-Hein,[24] Leonor Cerdá-Alberich,[25] Luis Martí-Bonmatí,[26] M Jorge Cardoso,[27] Maciej Bobowicz,[28] Mahsa Shabani,[29] Manolis Tsiknakis,[21] Maria A Zuluaga,[30] Marie-Christine Fritzsche,[31] Marina Camacho,[1] Marius George Linguraru,[32] Markus Wenzel,[9] Marleen De Bruijne,[33] Martin G Tolsgaard,[34] Melanie Goisauf,[35] Mónica Cano Abadía,[35] Nikolaos Papanikolaou,[36] Noussair Lazrak,[1] Oriol Pujol,[1] Richard Osuala,[1] Sandy Napel,[37] Sara Colantonio,[38] Smriti Joshi,[1] Stefan Klein,[33] Susanna Aussó,[39] Wendy A Rogers,[40] Zohaib Salahuddin,[41] Martijn P A Starmans[33]; on behalf of the FUTURE-AI Consortium

For numbered affiliations see end of the article

Correspondence to: K Lekadir karim.lekadir@ub.edu (ORCID 0000-0002-9456-1612)

Additional material is published online only. To view please visit the journal online.

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Accepted: 10 January 2025

Despite major advances in artificial intelligence (AI) research for healthcare, the deployment and adoption of AI technologies remain limited in clinical practice. This paper describes the FUTURE-AI framework, which provides guidance for the development and deployment of trustworthy AI tools in healthcare. The FUTURE-AI Consortium was founded in 2021 and comprises 117 interdisciplinary experts from 50 countries representing all continents, including AI scientists, clinical researchers, biomedical ethicists, and social scientists. Over a two year period, the FUTURE-AI guideline was established through consensus based on six guiding principles—fairness, universality, traceability, usability, robustness, and explainability. To operationalise trustworthy AI in healthcare, a set of 30 best practices were defined, addressing technical, clinical, socioethical, and legal dimensions. The recommendations cover the entire lifecycle of healthcare AI, from design, development, and validation to regulation, deployment, and monitoring.

SUMMARY POINTS

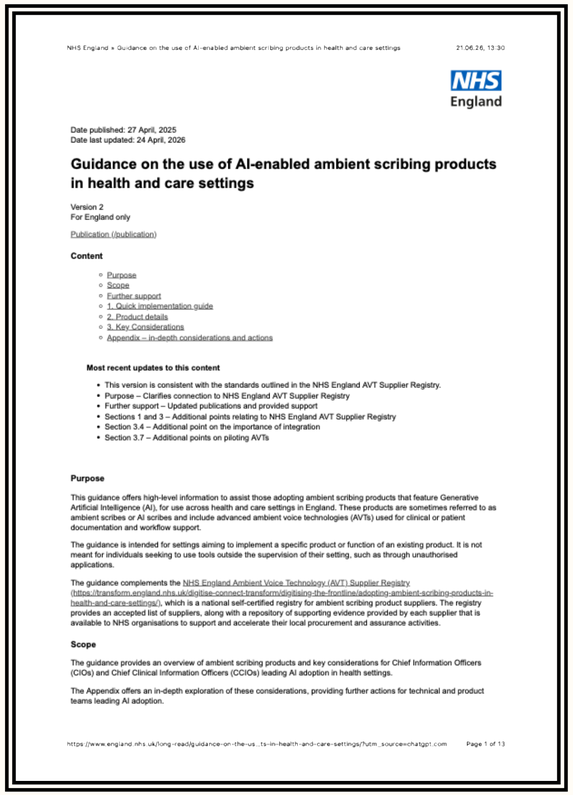
- Despite major advances in medical artificial intelligence (AI) research, clinical adoption of emerging AI solutions remains challenging owing to limited trust and ethical concerns
- The FUTURE-AI Consortium unites 117 experts from 50 countries to define international guidelines for trustworthy healthcare AI
- The FUTURE-AI framework is structured around six guiding principles: fairness, universality, traceability, usability, robustness, and explainability
- The guideline addresses the entire AI lifecycle, from design and development to validation and deployment, ensuring alignment with real world needs and ethical requirements
- The framework includes 30 detailed recommendations for building trustworthy and deployable AI systems, emphasising multistakeholder collaboration
- Continuous risk assessment and mitigation are fundamental, addressing biases, data variations, and evolving challenges during the AI lifecycle
- FUTURE-AI is designed as a dynamic framework, which will evolve with technological advancements and stakeholder feedback

Introduction

In the field of healthcare, artificial intelligence (AI)—that is, algorithms with the ability to self-learn logic—and data interactions have been increasingly used to develop computer aided models, for example, disease diagnosis, prognosis, prediction of therapy response or survival, and patient stratification.[1] Despite major advances, the deployment and adoption of AI technologies remain limited in real world clinical practice. In recent years, concerns have been raised about the technical, clinical, ethical, and societal risks associated with healthcare AI.[2,3] In particular, existing research has shown that AI tools in healthcare can be prone to errors and patient harm, biases and increased health inequalities, lack of transparency and accountability, as well as data privacy and security breaches.[4–8]

To increase adoption in the real world, it is essential that AI tools are trusted and accepted by patients, clinicians, health organisations, and authorities. However, there is an absence of clear, widely accepted guidelines on how healthcare AI tools should be designed, developed, evaluated, and deployed to be trustworthy—that is, technically robust, clinically safe,

Quelle: BMJ 2025;388:e081554



NHS England

Date published: 27 April, 2025
Date last updated: 24 April, 2026

Guidance on the use of AI-enabled ambient scribing products in health and care settings

Version 2
For England only

Publication (/publication)

Content

- Purpose
- Scope
- Further support
- 1. Quick implementation guide
- 2. Product details
- 3. Key Considerations
- Appendix – in-depth considerations and actions

Most recent updates to this content

- This version is consistent with the standards outlined in the NHS England AVT Supplier Registry.
- Purpose – Clarifies connection to NHS England AVT Supplier Registry
- Further support – Updated publications and provided support
- Sections 1 and 3 – Additional points relating to NHS England AVT Supplier Registry
- Section 3.4 – Additional point on the importance of integration
- Section 3.7 – Additional points on piloting AVTs

Purpose

This guidance offers high-level information to assist those adopting ambient scribing products that feature Generative Artificial Intelligence (AI), for use across health and care settings in England. These products are sometimes referred to as ambient scribes or AI scribes and include advanced ambient voice technologies (AVTs) used for clinical or patient documentation and workflow support.

The guidance is intended for settings aiming to implement a specific product or function of an existing product. It is not meant for individuals seeking to use tools outside the supervision of their setting, such as through unauthorised applications.

The guidance complements the NHS England Ambient Voice Technology (AVT) Supplier Registry (https://transform.england.nhs.uk/digitise-connect-transform/digitising-the-frontline/adopting-ambient-scribing-products-in-health-and-care-settings/), which is a national self-certified registry for ambient scribing product suppliers. The registry provides an accepted list of suppliers, along with a repository of supporting evidence provided by each supplier that is available to NHS organisations to support and accelerate their local procurement and assurance activities.

Scope

The guidance provides an overview of ambient scribing products and key considerations for Chief Information Officers (CIOs) and Chief Clinical Information Officers (CCIOs) leading AI adoption in health settings.

The Appendix offers an in-depth exploration of these considerations, providing further actions for technical and product teams leading AI adoption.

Quelle: NHS, https://www.england.nhs.uk/long-read/guidance-on-the-use-of-ai-enabled-ambient-scribing-products-in-health-and-care-settings



JGIM

POSITION PAPER

Recommendations for Clinicians, Technologists, and Healthcare Organizations on the Use of Generative Artificial Intelligence in Medicine: A Position Statement from the Society of General Internal Medicine

Byron Crowe, MD[1,2], Shreya Shah, MD[3,4], Derek Teng, MD[1,2], Stephen P. Ma, MD PhD[19], Matthew DeCamp, MD PhD[5], Eric I. Rosenberg, MD, MSPH[6], Jorge A. Rodriguez, MD[2,7], Benjamin X. Collins, MD MS MA[8,22], Kathryn Huber, MD MS[9], Kyle Karches, MD PhD[10], Shana Zucker, MD MPH MS[11], Eun Ji Kim, MD MS MS[12], Lisa Rotenstein, MD MBA[13], Adam Rodman, MD MPH[1,2], Danielle Jones, MD[14], Ilana B. Richman, MD, MHS[15], Tracey L. Henry, MD, MPH, MS[14], Diane Somlo, MD MBA[2,16], Samantha I. Pitts, MD MPH[17], Jonathan H. Chen, MD PhD[18,19,20], and Rebecca G. Mishuris, MD MS MPH[2,7,21]

[1]Division of General Internal Medicine, Beth Israel Deaconess Medical Center, Boston, MA, USA. [2]Harvard Medical School, Boston, MA, USA. [3]Department of Medicine, Stanford University, Palo Alto, CA, USA. [4]Division of Primary Care and Population Health, Stanford Healthcare AI Applied Research Team, Stanford University School of Medicine, Palo Alto, CA, USA. [5]Department of Medicine, University of Colorado, Aurora, CO, USA. [6]Division of General Internal Medicine, Department of Medicine, University of Florida College of Medicine, Gainesville, FL, USA. [7]Division of General Internal Medicine, Brigham and Women's Hospital, Boston, MA, USA. [8]Division of General Internal Medicine and Public Health, Vanderbilt University Medical Center, Nashville, TN, USA. [9]Department of Internal Medicine, Kaiser Permanente Denver, CO; School of Medicine, University of Colorado, Aurora, CO, USA. [10]Department of Internal Medicine, Saint Louis University, Saint Louis, MO, USA. [11]Department of Internal Medicine, University of Miami Miller School of Medicine, Jackson Memorial Hospital, Miami, FL, USA. [12]Northwell Health, New Hyde Park, NY, USA. [13]Divisions of General Internal Medicine and Clinical Informatics, Department of Medicine, University of California at San Francisco, San Francisco, CA, USA. [14]Division of General Internal Medicine, Emory University School of Medicine, Atlanta, GA, USA. [15]Section of General Internal Medicine, Yale School of Medicine, New Haven, CT, USA. [16]Department of Medicine, Massachusetts General Hospital, Boston, MA, USA. [17]Division of General Internal Medicine, Johns Hopkins University School of Medicine, Baltimore, MD, USA. [18]Stanford Center for Biomedical Informatics Research, Stanford, CA, USA. [19]Division of Hospital Medicine, Stanford, CA, USA. [20]Clinical Excellence Research Center, Stanford, CA, USA. [21]Digital, Mass General Brigham, Somerville, MA, USA. [22]Department of Biomedical Informatics, Vanderbilt University, Nashville, TN, USA*

ABSTRACT

Generative artificial intelligence (generative AI) is a new technology with potentially broad applications across important domains of healthcare, but serious questions remain about how to balance the promise of generative AI against unintended consequences from adoption of these tools. In this position statement, we provide recommendations on behalf of the Society of General Internal Medicine on how clinicians, technologists, and healthcare organizations can approach the use of these tools. We focus on three major domains of medical practice where clinicians and technology experts believe generative AI will have substantial immediate and long-term impacts: clinical decision-making, health systems optimization, and the patient-physician relationship. Additionally, we highlight our most important generative AI ethics and equity considerations for these stakeholders. For clinicians, we recommend approaching generative AI similarly to other important biomedical advancements, critically appraising its evidence and utility and incorporating it thoughtfully into practice. For technologists developing generative AI for healthcare applications, we recommend a major frameshift in thinking away from the expectation that clinicians will "supervise" generative AI. Rather, these organizations and individuals should hold themselves and their technologies to the same set of high standards expected of the clinical workforce and strive to design high-performing, well-studied tools that improve care and foster the therapeutic relationship, not simply those that improve efficiency or market share. We further recommend deep and ongoing partnerships with clinicians and patients as necessary collaborators in this work. And for healthcare organizations, we recommend pursuing a combination of both incremental and transformative change with generative AI, directing resources toward both endeavors, and avoiding the urge to rapidly displace the human clinical workforce with generative AI. We affirm that the practice of medicine remains a fundamentally human endeavor which should be enhanced by technology, not displaced by it.

KEY WORDS: clinical practice; artificial intelligence; healthcare technology

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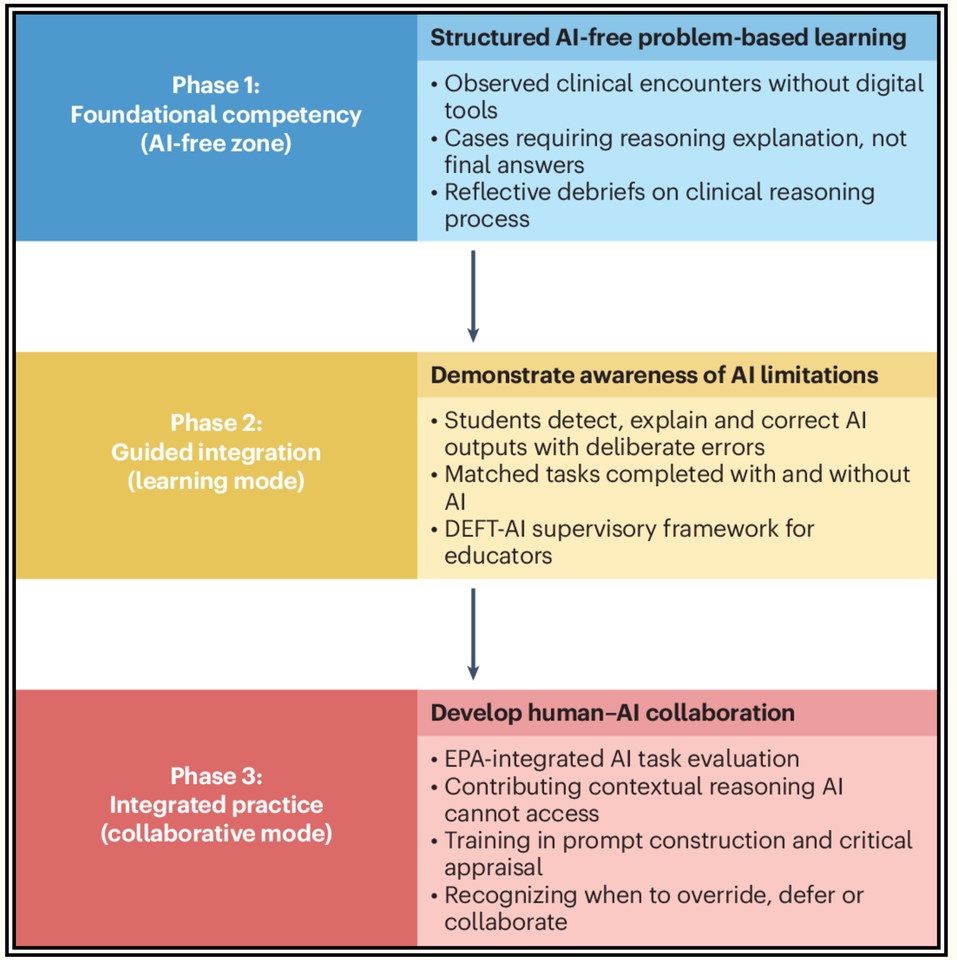
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Published online November 12, 2024

Healthcare Organizations on the Use of Generative Artificial Intelligence in Medicine: A Position Statement from the Society of General Internal Medicine. J GEN INTERN MED 40, 694–702 (2025).

Gute Ausbildung

“KI kann uns nicht das Verstehen abnehmen”

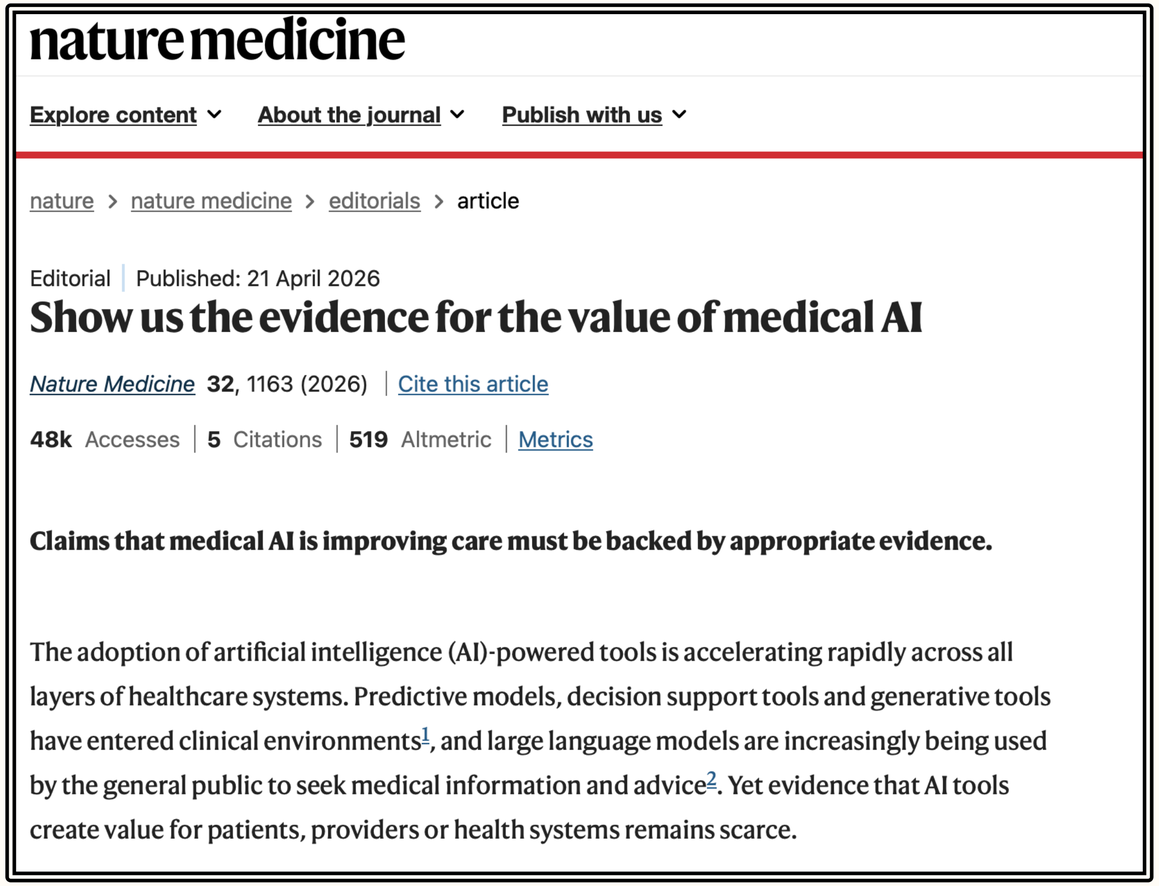
- Andrej Karpathy



Quelle: Ke, Y., Jin, L., Ong, J.C.L. et al. AI-induced never-skilling in medical education. Nat Med 32, 1997–2006 (2026).

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Show us the evidence for the value of medical AI

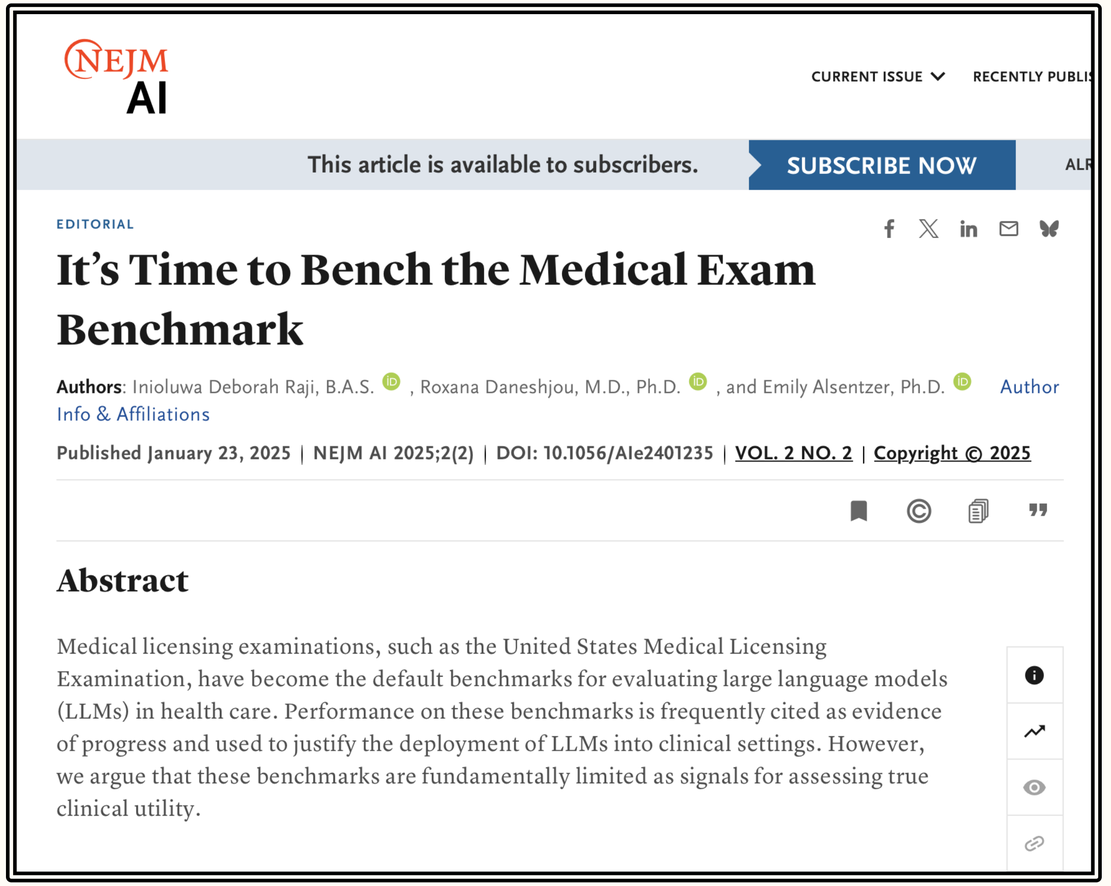
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Claims that medical AI is improving care must be backed by appropriate evidence.

The adoption of artificial intelligence (AI)-powered tools is accelerating rapidly across all layers of healthcare systems. Predictive models, decision support tools and generative tools have entered clinical environments[1], and large language models are increasingly being used by the general public to seek medical information and advice[2]. Yet evidence that AI tools create value for patients, providers or health systems remains scarce.

Quelle: Show us the evidence for the value of medical AI. Nat Med 32, 1163 (2026)



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EDITORIAL

It's Time to Bench the Medical Exam Benchmark

Authors: Inioluwa Deborah Raji, B.A.S. , Roxana Daneshjou, M.D., Ph.D. , and Emily Alsentzer, Ph.D. Author Info & Affiliations

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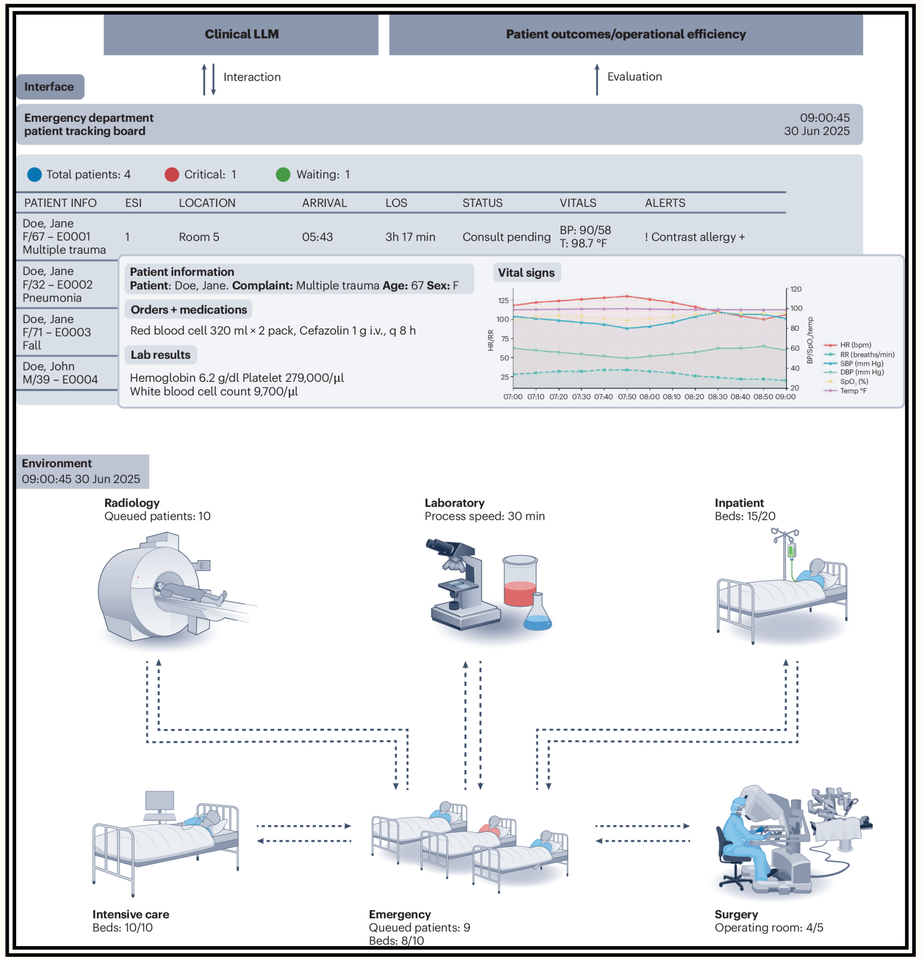
Abstract

Medical licensing examinations, such as the United States Medical Licensing Examination, have become the default benchmarks for evaluating large language models (LLMs) in health care. Performance on these benchmarks is frequently cited as evidence of progress and used to justify the deployment of LLMs into clinical settings. However, we argue that these benchmarks are fundamentally limited as signals for assessing true clinical utility.

Quelle: Raji I,Roxana Daneshjou R, Alsentzer E, It's Time to Bench the Medical Exam Benchmark, NEJM AI (2025)

Schrittweise Evaluation

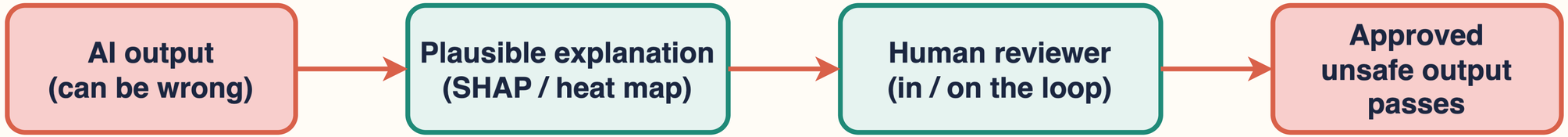
1. Statische Benchmarks
2. Retrospektive Validierung
3. Statische Vignettenstudien
4. Agentenbasierte Aufgaben in virtuellen EHR-Umgebungen
5. Dynamische Simulation mit aufeinander aufbauenden Entscheidungen
6. Prospektive Studien im Versorgungsalltag
7. Studien nach Einführung mit Daten aus der Versorgung
8. Begleitende ökonomische Evaluation



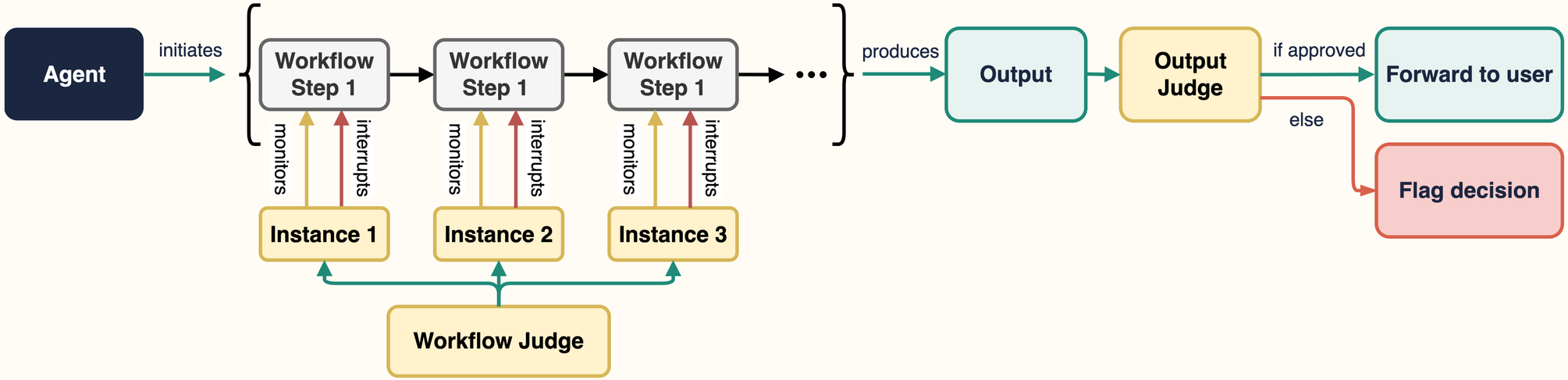
Quelle: Luo, L., Kim, S.E., Zhang, X. et al. A clinical environment simulator for dynamic AI evaluation. Nat Med 32, 820–827 (2026)

Automatisierte Aufsicht

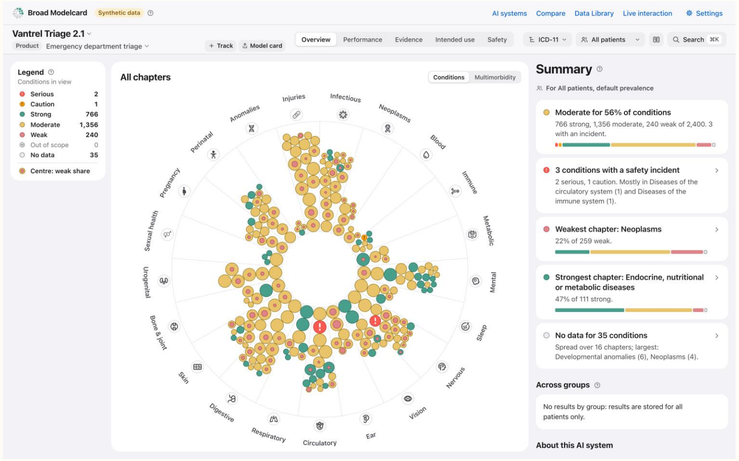
Manuelle Aufsicht

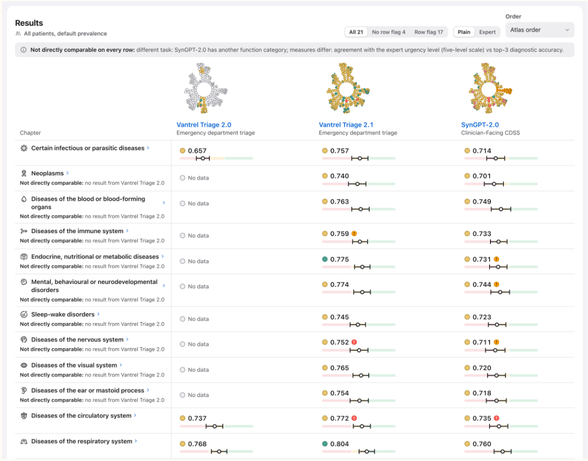


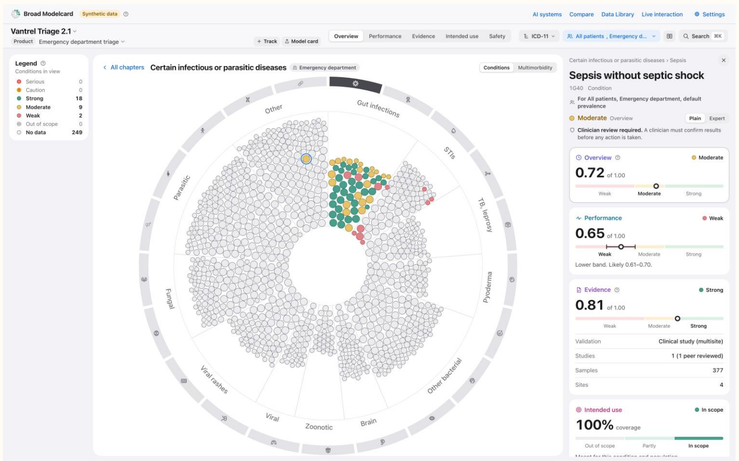
Agentische Aufsicht

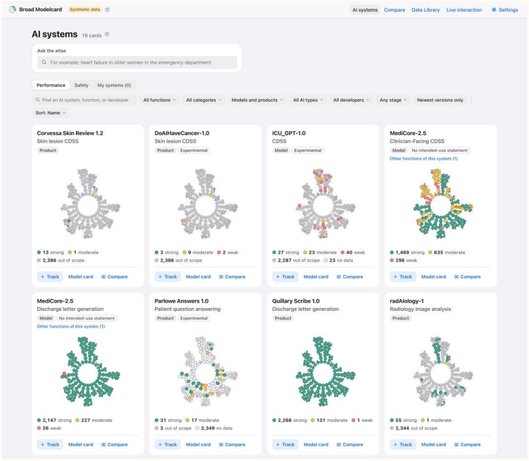


Broad Model Cards









Einsatzbereich
Aufgabe, Population, Setting

Evidenz
Ergebnisse, Quellen, Lücken

Version
Modell und Werkzeuge

Broad Model Cards. Prototyp mit synthetischen Daten.

Vielen Dank.

